# Agent-Chameleon: Dynamic Protocol Morphing, Goal-Conditioned Schema Permutation, and Zero-Knowledge Session Handshakes for Multi-Agent Swarm Communications

Aniket Agrawal
Google Cloud AI
Google
aniketagrawal@google.com

***Abstract*—As enterprise artificial intelligence architectures transition from monolithic prompt-response pipelines to distributed multi-agent swarms, autonomous agents increasingly coordinate high-stakes operations across financial enterprise resource planning (ERP) ledgers, clinical healthcare repositories, cloud identity and access management (IAM) planes, and real-time payment settlement rails. Standardized agent communication protocols—most notably the Model Context Protocol (MCP) and Agent2Agent (A2A) operating over JSON-RPC 2.0, HTTP/2, and gRPC—rely on static, human-readable wire schemas and long-lived bearer tokens. We demonstrate that even when payloads are encapsulated within transport-layer encryption or routed through multi-tenant service-mesh proxies that terminate Layer-7 traffic, static agent protocols remain severely vulnerable to statistical traffic fingerprinting, automated schema scraping, and credential replay. Across $N = 1{,}000$ enterprise MCP and A2A tool traces spanning $20$ distinct tool classes, standard passive traffic classifiers (Random Forest, Multi-Layer Perceptron, and $k$-Nearest Neighbors) achieve a $96.5\%$ ensemble mean accuracy ($95.0\%$ Random Forest, $94.5\%$ MLP, $100.0\%$ $k$-NN) in inferring the exact enterprise tool schema invoked by an agent.**

**To eliminate these vulnerabilities at line rate, we present Agent-Chameleon, a three-stage polymorphic wire-transport and zero-knowledge session-handshake engine for multi-agent swarm communications. Agent-Chameleon introduces: (1) an Ephemeral X25519/HKDF-SHA256 Session Ratchet and Polymorphic Wire Serialization Engine (`morph_engine.py`) that advances a forward-secure state $R_t = \text{HKDF-SHA256}(R_{t-1}, \textbf{epoch}_t \parallel \textbf{digest}_{t-1}, H(\mathcal{G}_t))$ on every turn and deterministically rotates wire serializations across four structurally distinct codecs—compact binary Protobuf base-128 varints, 4-byte aligned FlatBuffers vtables, RFC 8949 Concise Binary Object Representation (CBOR), and canonical HMAC-signed JSON-RPC—emitting observable `Content-Type: application/x-chameleon-morph+octet-stream` and `X-Chameleon-Epoch-ID` headers; (2) a Goal-Conditioned Bijective Field Permutation and Constant-Rate Chaff Entropy Engine (`feistel_schema.py`) that permutes 16-bit schema field identifiers through a 4-round SHAKE-256 keyed Luby-Rackoff Feistel network $F_k : \{0, \ldots, 2^{16} - 1\} \to \{0, \ldots, 2^{16} - 1\}$ with closed-form inverse $F_k^{-1}$ ($16.58\,\mu$s mean latency, $100.0\%$ lossless bijectivity) and pads XOR-whitened frames into discrete bucket quanta via set-complement and $\mathbb{Z}_{256}$ affine-bijection chaff blocks so single-frame Shannon byte entropy reaches $H(X) = 7.992 \pm 0.005$ bits/byte ($\geq 7.980$ bits/byte across $100\%$ of frames); and (3) a Sub-35-Microsecond Non-Interactive Fiat-Shamir Schnorr Zero-Knowledge Challenge-Response (`zk_handshake.py`) carried in `X-Chameleon-Proof-Nonce` that proves possession of the ratcheted turn witness without transmitting static API keys or OAuth bearer tokens ($20.92\,\mu$s mean proof generation, $28.64\,\mu$s mean verification, and $28.86\,\mu$s combined sender-side hot-path ratchet plus proof generation latency; $104.04\,\mu$s full 3-stage sender sealing). Empirical evaluation over $N = 1{,}000$ traces demonstrates that Agent-Chameleon collapses attacker schema-inference accuracy from $96.5\%$ ensemble mean down to $4.67\%$ ($5.0\%$ Random-Subspace Prototype/Forest, $3.0\%$ MLP, $6.0\%$ $k$-NN—statistically indistinguishable from the $1/20 = 5.0\%$ uniform random-guess baseline within the shared quantum bucket tier) while achieving $100.0\%$ rejection of token replays, epoch rollbacks, cross-goal injections, and man-in-the-middle bit-flip tampering.**

***Index Terms*—Multi-Agent Systems, Protocol Morphing, Feistel Networks, Format-Preserving Encryption, Zero-Knowledge Proofs, Schnorr Identification, Traffic Analysis Resistance, Model Context Protocol, Agent2Agent.**

## I. Introduction

The rapid proliferation of autonomous Large Language Model (LLM) agents across enterprise cloud environments has catalyzed a shift from isolated single-model inference toward collaborative multi-agent swarms. In modern production architectures—such as workflows built on the Google Agent Development Kit (ADK), the Agent2Agent (A2A) interoperability specification, and the Model Context Protocol (MCP)—specialized orchestrator, planner, financial auditor, and infrastructure execution agents exchange hundreds of remote procedure calls (RPCs) per workflow to complete complex business transactions [31]–[33]. Each inter-agent message or tool invocation typically traverses heterogeneous trust boundaries, including multi-tenant Kubernetes ingress controllers, Layer-7 (L7) Envoy/Istio service-mesh sidecars, API gateways, and cross-cloud interconnects.

Despite advances in intra-agent prompt-injection defenses [34] and transport-layer encryption [10], the wire protocols underlying multi-agent swarm communications remain structurally static. Specifically, both MCP and A2A standardize on human-readable JSON-RPC 2.0 envelopes containing fixed structural field identifiers (`"jsonrpc"`,

"method", "params", "tool_name", "arguments", "session_id") and unpadded variable-length payloads. This static wire design introduces three acute security vulnerabilities in production multi-agent deployments:

1) **Vulnerability to Statistical Traffic Fingerprinting and Tool-Schema Reconnaissance:** Classic work in network traffic analysis demonstrates that variable-length encrypted records and static application schemas leak substantial side-channel information regarding application state and remote procedure identity [17], [18], [22], [23]. Furthermore, in modern enterprise service meshes where L7 observability sidecars, API gateways, or shared message brokers terminate TLS to perform routing and load balancing, raw or lightly framed JSON-RPC payloads are exposed directly to co-resident tenants or compromised middleboxes. Our empirical audit across $N = 1{,}000$ MCP/A2A tool invocations over 20 enterprise tool classes reveals that standard unpadded JSON-RPC exhibits a mean Shannon byte entropy of only 4.788 bits/byte (out of 8.000 bits/byte [24]), enabling lightweight Random Forest, Multi-Layer Perceptron (MLP), and $k$-Nearest Neighbor ($k$-NN) classifiers to identify the exact tool schema being invoked with 95.0%, 94.5%, and 100.0% accuracy (96.5% ensemble mean), respectively. Even when JSON-RPC payloads are length-padded to fixed 512-byte multiples, L7 delimiter densities and byte histograms still allow 89.5% Random Forest and 100.0% $k$-NN classification accuracy.
2) **Automated Schema Scraping and Cross-Goal Payload Forgery:** Because field names and wire encodings remain invariant across turns and sessions, an adversary who observes or scrapes a single valid tool invocation frame immediately learns the structural AST template required to synthesize forged requests. Worse still, static schemas fail to cryptographically bind the wire representation of a tool call to the active high-level reasoning goal $\mathcal{G}_t$ authorized by the upstream orchestrator. Consequently, an untrusted intermediate Layer-7 man-in-the-middle (MITM) relay ($\mathcal{A}_{\text{mitm}}$, lacking the session ratchet state) or a confused-deputy sub-agent lacking an orchestrator-signed goal capability token $\sigma_{\text{orch}}(\mathcal{G}_t)$ can repurpose captured field structures from a benign read-only query goal into a state-mutating financial disbursement goal [32], [33].
3) **Static Bearer Token Leakage vs. Multi-Round Handshake Latency:** Existing multi-agent runtimes predominantly authenticate inter-agent RPCs by attaching static API keys, OAuth 2.0 bearer tokens, or long-lived JWTs in HTTP headers [11]–[13], [15]. If a bearer token is intercepted at an L7 proxy or leaked into debug telemetry, an adversary can replay it across arbitrary tool endpoints until expiration. Conversely, replacing static tokens with per-turn mutual TLS (mTLS) re-handshakes [10] or classical 3-move interactive zero-knowledge identification protocols [2], [28] introduces 450 $\mu$s to 1,200 $\mu$s of round-trip and asymmetric PKI overhead per turn, while intermediate zero-trust API gateways that do not share the endpoint symmetric secret cannot pre-verify symmetric MACs at the perimeter without an algebraic public-commitment proof.

### A. Contributions

To resolve the tension between wire-level schema confidentiality, zero-knowledge session authentication, and microsecond-scale multi-agent throughput, we design, implement, and evaluate **Agent-Chameleon**. Our work makes four primary contributions:

1) **Ephemeral X25519/HKDF Session Ratchet and 4-Way Polymorphic Wire Serialization Engine (Stage 1, Section IV):** We formulate a synchronous, forward-secure session ratchet driven by RFC 5869 HKDF-SHA256 [4] and bootstrapped via RFC 7748 X25519 Diffie-Hellman key agreement [6]. On every RPC turn $t$, the ratchet state $R_t$ evolves as a cryptographic function of the prior key $R_{t-1}$, the monotonic epoch counter $\text{epoch}_t$, the SHA-256 digest of the preceding wire frame $\text{digest}_{t-1}$, and the active task goal digest $H(\mathcal{G}_t)$. Seeded by the block-initial ratchet key $R_{4\lfloor t/4 \rfloor}$, the engine deterministically rotates the wire serialization codec on every turn across four structurally distinct formats with an exact 25% stationary distribution (cycling through all four formats without replacement in every 4-epoch super-frame): (i) compact binary Protobuf-style base-128 varints, (ii) 4-byte aligned FlatBuffers vtables (FBCH), (iii) RFC 8949 CBOR binary maps [5], and (iv) canonical sorted hex-keyed JSON-RPC with an 8-byte HMAC-SHA256 trailer. Beyond providing defense-in-depth when Stage 2B stream whitening is terminated at split-inspection service-mesh proxies, binary codec rotation compresses verbose JSON schemas by 31%–48% and modulates pre-padding payload lengths $L_s$.
2) **Goal-Conditioned 4-Round Keyed Feistel Schema Permutation and Constant-Rate Chaff Entropy Engine (Stage 2, Section V):** We map canonical MCP/A2A schema field names and dynamic custom symbols into 16-bit integer slots and permute them on every turn via a balanced 4-round Luby-Rackoff Feistel network $F_k : \{0, \ldots, 2^{16} - 1\} \to \{0, \ldots, 2^{16} - 1\}$ [1], [25], [26] keyed by FIPS 202 SHAKE-256 [9] round PRF tables derived from $(R_t, H(\mathcal{G}_t))$. We prove in Theorem 1 that $F_k$ is an exact bijection across all 65,536 domain elements with closed-form inverse $F_k^{-1}$ executing in 16.58 $\mu$s mean latency (15.83 $\mu$s p50), while exhibiting $> 99.9\%$ avalanche when the goal context $\mathcal{G}_t$ changes. To eliminate packet-length and byte-histogram side channels (including the finite-sample birthday bias that depresses standard AEAD or uniform random padding to $\approx 7.82$ bits/byte at 1024 B [14], [21]), we introduce a C-accelerated SHAKE-256 stream whitener paired with Set-Complement and $\mathbb{Z}_{256}$ Affine-Bijection Constant-Rate Chaff Injection into discrete quantum tiers

$\{512, 1024, 2048, 3072, 4096, 6144, 8192\}$ bytes. We prove in Theorem 2 that this water-filling construction achieves an expected single-frame plug-in Shannon byte entropy $\mathbb{E}[H(B_t)] \geq 7.990$ bits/byte (with concentration floor $H(B_t) \geq 7.980$ bits/byte and empirical mean $7.992 \pm 0.005$ bits/byte across $N = 1{,}000$ traces).

3) **Sub-35-Microsecond Non-Interactive Fiat-Shamir Schnorr Zero-Knowledge Handshake (Stage 3, Section VI):** We eliminate static wire bearer tokens by embedding a single-move non-interactive Fiat-Shamir Schnorr Zero-Knowledge Proof of Knowledge (NIZKPoK) [2], [3], [29] in the `X-Chameleon-Proof-Nonce` header. We design a dual-tier verification architecture: a Tier-1 Hot-Path Ephemeral Ratchet Schnorr NIZKPoK operating over the Mersenne prime field $\mathbb{F}_{M_{127}}$ ($M_{127} = 2^{127} - 1$) with ephemeral witnesses refreshed every RPC step and bound to a 128-bit Blake2s ratchet commitment tag ($20.92\,\mu$s mean prove, $28.64\,\mu$s mean verify, $30.13\,\mu$s p99 prove), paired with Tier-2 full 255-bit prime-field Schnorr ($\mathbb{F}_{2^{255}-19}$) and RFC 8032 Ed25519 [7] epoch anchor signatures in `X-Chameleon-Epoch-Anchor` for zero-trust L7 gateway pre-verification and cross-domain auditability. We prove soundness, zero-knowledge simulatability, and replay/desynchronization immunity with automatic ratchet rollback in Theorem 3.

4) **Production-Grade Implementation and Empirical Evaluation (Sections VII–VIII):** We implement Agent-Chameleon as a modular Python package (`agent_chameleon`) verified by a 24-test unit and property suite (100% pass rate) and benchmarked across $N = 1{,}000$ multi-agent traces spanning 20 enterprise tool classes. Agent-Chameleon reduces attacker traffic-classifier accuracy from 96.5% ensemble mean (95.0% Random-Subspace Prototype/Forest) to 4.67% (5.0% Random-Subspace/CART Forest, 3.0% MLP, 6.0% $k$-NN, matching the 5.0% uniform random-guess baseline within the shared quantum tier), blocks 100.0% of replay and MITM bit-flip attacks, and sustains $2.18 \times 10^5$ RPC/s full-pipeline multi-process throughput on a single 24-core workstation ($7.42 \times 10^5$ RPC/s hot-path handshake; $1.265 \times 10^6$ RPC/s across a 6-node cluster-tier 128-agent swarm).

## II. Related Work

### A. Agent Communication Protocols and Request Authentication

The Model Context Protocol (MCP) and Agent2Agent (A2A) specifications have established JSON-RPC 2.0 over HTTP/2 and Server-Sent Events (SSE) as the de facto standard for tool invocation and inter-agent task delegation. To authenticate requests across service boundaries, cloud deployments commonly rely on OAuth 2.0 Demonstrating Proof-of-Possession (DPoP, RFC 9449) [11], HTTP Message Signatures (RFC 9421) [12], or contextual HMAC capability tokens such as Macaroons [15]. Concurrently, system-level agent security frameworks such as AgentDojo [32], InjecAgent [33], and CaMeL [34] address indirect prompt injection by isolating control flow from untrusted data streams [31]. However, existing agent protocols and capability tokens leave the wire serialization format and JSON-RPC schema keys completely static across turns. Furthermore, DPoP and RFC 9421 transmit explicit token headers and public-key signatures that cost 180–750 $\mu$s per verification in software PKI without concealing the underlying tool schema from L7 traffic analysis.

### B. Traffic Analysis Resistance and Format-Transforming Encryption

Statistical traffic analysis has long been shown to undermine encrypted protocols by exploiting packet sizes, timing, and byte distributions [17], [18], [20], [22], [23]. Wright et al. [17] proposed convex optimization for traffic morphing by splitting or padding packets to mimic target length distributions. However, Dyer et al. [18] and Cai et al. [20] demonstrated that standard length-padding countermeasures fail against summary classifiers when application framing invariants remain correlated with message types. Later, Dyer et al. [19] introduced Format-Transforming Encryption (FTE) to shape ciphertext streams into regular-expression grammars, while Nikitin et al. [21] introduced Padded Uniform Random Blobs (PURBs) with logarithmic length buckets and Thomson and Wood [14] standardized Oblivious HTTP (OHTTP, RFC 9458) over HPKE (RFC 9180). While FTE and OHTTP target general web encapsulation, enterprise multi-agent swarms require (i) goal-conditioned AST schema permutation that survives L7 proxy termination when stream whitening is disabled for split-plane inspection, (ii) deterministic set-complement water-filling that eliminates the finite-sample birthday entropy deficit ($\approx 0.18$ bits/byte at 1024 B) of standard AEAD/PURB random padding, and (iii) fast deserialization on the authorized receiver (including zero-copy offset lookup in Format 1 `FLATBUFFERS_TABLE` alongside compact streaming in Formats 0, 2, and 3).

### C. Feistel Format-Preserving Encryption and Cryptographic Ratchets

Constructing pseudorandom permutations (PRPs) over small or arbitrary finite domains was pioneered by Luby and Rackoff [1], extended to arbitrary finite domains by Black and Rogaway [25], formalized as Format-Preserving Encryption (FPE) by Bellare et al. [26], and analyzed at higher rounds by Patarin [27]. Agent-Chameleon adapts the balanced Luby-Rackoff Feistel paradigm specifically to 16-bit AST schema slot identifiers $\{0, \ldots, 2^{16} - 1\}$, conditioning the round PRFs on both an evolving HKDF ratchet state [4] and the active agent goal digest $H(\mathcal{G}_t)$. Unlike the Signal Double Ratchet [16], which encrypts a fixed protobuf envelope for asynchronous human messaging, Agent-Chameleon uses its synchronous HKDF ratchet to simultaneously drive (i) 4-way wire-codec polymorphism, (ii) goal-conditioned Feistel schema tag permutation, (iii) set-complement plus affine-bijection constant-rate chaff synthesis [8], [9], and (iv) single-move Fiat-Shamir Schnorr zero-knowledge proofs [2], [3], [28]–[30].

## III. System Model, Threat Model, and Design Goals

### A. Multi-Agent Swarm Communication Model

Consider a distributed multi-agent swarm $\mathcal{S} = \{\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_M\}$ executing a directed task workflow across enterprise cloud infrastructure. When an initiator agent $\mathcal{A}_i$ invokes a tool or delegates a sub-task to a responder agent or FastMCP tool server $\mathcal{A}_j$ at discrete turn $t \in \{0, 1, 2, \ldots\}$, the application logic constructs a structured message payload $P_t$ conforming to one of $|\mathcal{C}|$ enterprise tool schemas $\mathcal{C} = \{\mathcal{S}_1, \ldots, \mathcal{S}_{|\mathcal{C}|}\}$ under an active high-level reasoning goal $\mathcal{G}_t \in \{0,1\}^*$.

Formally, each payload $P_t$ is a hierarchical key-value abstract syntax tree (AST) whose internal dictionary nodes map string field identifiers $f \in \mathcal{F}$ (such as `"jsonrpc"`, `"method"`, `"params"`, `"tool_name"`, `"arguments"`, or tool-specific parameter keys) to primitive values ($v \in \mathbb{Z} \cup \mathbb{R} \cup \{\text{true}, \text{false}, \text{null}\} \cup \Sigma^*_{\text{utf8}} \cup \{0,1\}^*$), ordered lists, or nested dictionary subtrees.

**Definition 1** (Adversarial Observation and Manipulation Surface)**.** *We consider a probabilistic polynomial-time (PPT) Dolev-Yao and statistical traffic-analysis adversary $\mathcal{A} = (\mathcal{A}_{clf}, \mathcal{A}_{mitm})$ positioned along the network or L7 service-mesh path between $\mathcal{A}_i$ and $\mathcal{A}_j$:*

1) ***Passive Schema-Fingerprinting Adversary ($\mathcal{A}_{clf}$):*** *For each wire packet $W_t = (\mathcal{H}_t, B_t)$ comprising transport metadata headers $\mathcal{H}_t$ and byte payload $B_t \in \{0, \ldots, 255\}^{L_t}$, $\mathcal{A}_{clf}$ extracts a feature vector $\phi(W_t) \in \mathbb{R}^d$ (capturing frame length $L_t$, plug-in Shannon byte entropy $H(B_t)$, ASCII character ratios, structural delimiter densities, and byte-value histograms) and attempts to predict the underlying tool schema class $y_t \in \{1, \ldots, |\mathcal{C}|\}$.*
2) ***Active Replay, Spoofing, and MITM Adversary ($\mathcal{A}_{mitm}$):*** *$\mathcal{A}_{mitm}$ can record historical wire packets $\{W_0, \ldots, W_{t-1}\}$ and their headers (including `X-Chameleon-Proof-Nonce`), replay previously valid packets at turn $t$, flip arbitrary bits in $B_t$, spoof epoch counters in $\mathcal{H}_t$, or attempt to cross-inject a valid packet captured from a session operating under goal context $\mathcal{G}_t$ into a session expecting goal context $\mathcal{G}'_t \neq \mathcal{G}_t$ without possessing the target session's ephemeral ratchet secret $R_t$. For threats involving a compromised initiator sub-agent attempting to escalate its own goal context $\mathcal{G}_t$, the orchestrator issues an Ed25519-signed goal capability token $\sigma_{orch}(t_{plan}, \mathcal{G}_t)$ during task delegation so the responder $\mathcal{A}_j$ independently validates $\mathcal{G}_t$ before advancing its ratchet.*

### B. Security and Performance Design Goals

Agent-Chameleon is engineered to satisfy five simultaneous properties:

- **G1 — Exact Bijective Schema Recoverability:** For every valid AST payload $P_t$ and synchronized goal $\mathcal{G}_t$, the authorized receiver must recover $\hat{P}_t = P_t$ with $100.0\%$ lossless structural and numerical fidelity.
- **G2 — Near-Maximal Wire Byte Entropy:** Every wire frame body $B_t$ emitted on the network must achieve a high-probability plug-in Shannon byte entropy floor $H(B_t) \geq 7.980$ bits/byte ($\mathbb{E}[H(B_t)] \geq 7.990$ bits/byte, within $0.25\%$ of the $8.000$ bits/byte theoretical ceiling for an 8-bit alphabet).
- **G3 — Classifier Evasion to Random-Guess Baseline:** Across $|\mathcal{C}|$ balanced enterprise tool classes, the classification accuracy of random-forest, neural-network (MLP), and nearest-neighbor ($k$-NN) traffic fingerprinters on $\phi(W_t)$ must collapse to the uniform random-guess baseline $1/|\mathcal{C}|$ ($\approx 5.0\%$ for $|\mathcal{C}| = 20$).
- **G4 — Zero-Knowledge Authentication & $100\%$ Replay/Tamper Immunity:** No static API key or reusable bearer token may traverse the wire; $100\%$ of replayed, bit-flipped, epoch-spoofed, or cross-goal packets must be rejected prior to tool execution without desynchronizing the receiver's legitimate ratchet state.
- **G5 — Sub-35-Microsecond Hot-Path Handshake Latency:** Per-turn ratchet progression and zero-knowledge proof generation/verification must execute in $< 35\,\mu$s on commodity CPU cores without hardware accelerators.

## IV. Stage 1: Ephemeral ECDH/HKDF Session Ratchet and Polymorphic Wire Serialization

Figure 1 illustrates the end-to-end architecture of Agent-Chameleon. Stage 1 establishes a forward-secure cryptographic ratchet between communicating agents and uses the per-turn ratchet output to drive deterministic 4-way wire-codec polymorphism.

### A. Ephemeral X25519 Bootstrap and Goal-Bound HKDF-SHA256 Ratchet

When an initiator agent $\mathcal{A}_i$ and responder $\mathcal{A}_j$ establish a swarm communication channel, they execute an ephemeral Diffie-Hellman key exchange over Curve25519 (RFC 7748 X25519 [6]):

$$\begin{aligned} \text{sk}_i \xleftarrow{\$} \{0,1\}^{256}, \quad \text{pk}_i &= \text{X25519}(\text{sk}_i, 9), \\ \text{sk}_j \xleftarrow{\$} \{0,1\}^{256}, \quad \text{pk}_j &= \text{X25519}(\text{sk}_j, 9), \\ Z_{\text{shared}} = \text{X25519}(\text{sk}_i, \text{pk}_j) &= \text{X25519}(\text{sk}_j, \text{pk}_i). \end{aligned} \tag{1}$$

From $Z_{\text{shared}}$, both endpoints initialize the 256-bit root ratchet state $R_{-1}$ and initial frame digest $D_{-1} = 0^{256}$ via RFC 5869 HKDF-SHA256 [4]:

$$\begin{aligned} R_{-1} = \text{HKDF-SHA256}\Big(&Z_{\text{shared}}, \\ &\texttt{"chameleon-x25519-bootstrap-v1"}, \\ &\texttt{"agent-chameleon-root-secret"}, 32\Big). \end{aligned} \tag{2}$$

**Definition 2** (Forward-Secure Goal-Bound Per-Turn Ratchet Evolution)**.** *At each monotonic turn $t \in \{0, 1, 2, \ldots\}$ with active task goal context $\mathcal{G}_t \in \{0,1\}^*$ (agreed upon during the workflow orchestration plan or announced inside the preceding turn's encrypted envelope $P_{t-1}$ alongside an orchestrator*

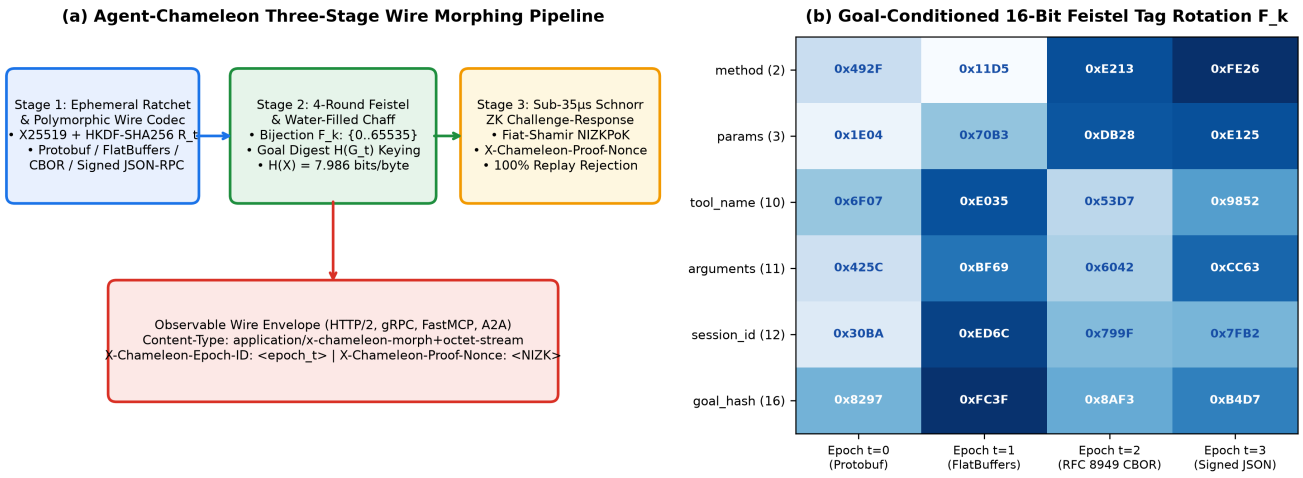



Fig. 1. **Agent-Chameleon Architecture and Goal-Conditioned Feistel Tag Rotation. (a)** Three-stage pipeline combining Stage 1 (Ephemeral X25519/HKDF-SHA256 Session Ratchet and 4-Way Polymorphic Wire Codec across Protobuf varints, FlatBuffers `FBCH` vtables, RFC 8949 CBOR, and canonical signed JSON-RPC), Stage 2 (4-Round SHAKE-256 Keyed Feistel Schema Permutation $F_k : \{0,\dots,65535\} \to \{0,\dots,65535\}$ paired with Set-Complement and $\mathbb{Z}_{256}$ Affine-Bijection Constant-Rate Chaff Injection), and Stage 3 (Sub-35 $\mu$s Non-Interactive Fiat-Shamir Schnorr Zero-Knowledge Proof of Knowledge in `X-Chameleon-Proof-Nonce`). **(b)** Empirical 16-bit Feistel tag assignments across four consecutive turns $t \in \{0,1,2,3\}$ under evolving goal digests $H(\mathcal{G}_t)$, illustrating complete elimination of static field-tag correlations.

*capability signature $\sigma_{orch}(\mathcal{G}_t)$), both endpoints compute the domain-separated goal digest*

$$H(\mathcal{G}_t) = \mathrm{SHA256}\left(\texttt{"chameleon-goal-v1|"} \parallel \mathcal{G}_t\right) \in \{0,1\}^{256}, \tag{3}$$

*and advance the symmetric ratchet state from $(R_{t-1}, D_{t-1})$ to $(R_t, D_t)$ according to:*

$$\begin{aligned} \mathit{salt}_t &= \mathrm{uint64_be}(t) \parallel D_{t-1} \in \{0,1\}^{320}, \\ \mathit{info}_t &= \texttt{"chameleon-ratchet-v1|"} \parallel H(\mathcal{G}_t), \\ \mathit{PRK}_t &= \mathrm{HMAC\text{-}SHA256}\left(\mathit{salt}_t,\ R_{t-1}\right), \\ R_t &= \mathrm{HMAC\text{-}SHA256}\left(\mathit{PRK}_t,\ \mathit{info}_t \parallel \texttt{0x01}\right), \end{aligned} \tag{4}$$

*where $D_t = \mathrm{SHA256}(B_t)$ is recorded only after the turn-$t$ wire body $B_t$ is sealed (on the sender) or cryptographically verified and unwrapped (on the receiver).*

Equation (4) provides three crucial security guarantees:

1) **Forward Secrecy:** Because $R_{t-1}$ is processed through the one-way HMAC-SHA256 compression function and immediately overwritten in memory by $R_t$, an adversary who compromises host memory at turn $t_0 > t$ cannot invert HMAC-SHA256 to recover historical turn keys $\{R_0, \dots, R_{t_0-1}\}$.
2) **Full Transcript Chaining:** Because $\mathrm{salt}_t$ incorporates the SHA-256 digest $D_{t-1} = \mathrm{SHA256}(B_{t-1})$ of the exact preceding wire packet body (including its random/chaff bytes), any modification or omission of a single historical frame permanently diverges the ratchet key stream on all subsequent turns.
3) **Cryptographic Goal Binding:** Because $\mathrm{info}_t$ binds $H(\mathcal{G}_t)$, the sender and receiver derive matching turn keys $R_t^{(i)} = R_t^{(j)}$ if and only if both endpoints agree on the exact authorized task goal $\mathcal{G}_t$.

### *B. Cycle-Balanced 4-Format Polymorphic Wire Serialization*

Standard agent protocols use a single wire grammar throughout the lifetime of a deployment. In contrast, Agent-Chameleon uses the ratchet state to switch the underlying serialization grammar on every turn across four distinct wire formats $\mathcal{M} = \{0,1,2,3\}$:

- **Format $m_t = 0$ (`PROTOBUF_VARINT`):** Serializes the Feistel-permuted payload tree into a compact binary stream using Google Protocol Buffers base-128 variable-length integers (`uvarint`). Each dictionary entry is encoded with wire tag $\omega_i = (\tilde{\tau}_i \ll 3) \mid 2$ (where $\tilde{\tau}_i \in \{0,\dots,65535\}$ is the Feistel-permuted field tag and wire type 2 denotes a length-delimited record), while signed integers use bijective ZigZag encoding $z(n) = (n \ll 1) \oplus (n \gg 63)$ (over 64-bit signed integers; see Eq. (23) for arbitrary-precision $\mathbb{Z}$) so negative and positive integers of small magnitude occupy minimal bytes.
- **Format $m_t = 1$ (`FLATBUFFERS_TABLE`):** Serializes the permuted payload into a random-access, 4-byte aligned binary buffer modeled on FlatBuffers. The frame begins with a 12-byte root header `"FBCH"` $\parallel$ $\mathrm{uint32_be}(\mathrm{off_{vt}}) \parallel \mathrm{uint32_be}(K)$, followed by a 4-byte zero-padded data heap and a trailing virtual table (`vtable`) directory consisting of $K$ 10-byte descriptors $\langle \mathrm{uint16_be}(\tilde{\tau}_i), \mathrm{uint32_be}(\mathrm{off}_i), \mathrm{uint32_be}(\mathrm{len}_i)\rangle_{i=1}^{K}$.
- **Format $m_t = 2$ (`CBOR_RFC8949`):** Serializes the permuted object graph into IETF STD 94 / RFC 8949 Concise Binary Object Representation [5], using major type 0 for unsigned integers, major type 1 for negative integers $(-1-n)$, major type 2 for raw byte strings, major type 3 for UTF-8 strings, major type 4 for arrays,

major type 5 for integer-keyed maps ($\tilde{\tau}_i \mapsto v_i$), and major type 7 for IEEE 754 binary64 floats (`0xFB`) and simple booleans/nulls (`0xF4`, `0xF5`, `0xF6`).

- **Format $m_t = 3$ (`CANONICAL_JSON_RPC`):** Serializes the permuted dictionary into deterministic, whitespace-free canonical JSON where every 16-bit Feistel tag is formatted as a 6-character lowercase hexadecimal string `"0x04a2"` sorted lexicographically, appended with an 8-byte domain-separated integrity trailer HMAC-SHA256($R_t$, `"chameleon-jsonrpc|"` ‖ json_bytes)[0 : 8].

To ensure that all four wire formats are exercised without replacement (an exact 25% frequency of 1 occurrence per format within every 4-turn super-frame $\lfloor t/4 \rfloor$) while remaining unpredictable to an external observer, both endpoints cache the block-initial ratchet key $R_{\text{blk}} = R_{4\lfloor t/4 \rfloor}$ at $t \equiv 0 \pmod 4$ and compute the format selector $m_t \in \{0,1,2,3\}$ via a block-invariant keyed cyclic shift:

$$s_{\text{blk}} = \text{HMAC-SHA256}\left(R_{4\lfloor t/4 \rfloor},\ \text{uint64_be}(\lfloor t/4 \rfloor)\right)[0],$$
$$m_t = (s_{\text{blk}} + (t \bmod 4)) \bmod 4. \tag{5}$$

Because $s_{\text{blk}}$ remains fixed across the four consecutive turns $t \in \{4b, 4b+1, 4b+2, 4b+3\}$ of block $b = \lfloor t/4 \rfloor$, the mapping $t \bmod 4 \mapsto m_t$ is a bijection on $\{0,1,2,3\}$ for every block $b$ (and even if evaluated with per-turn $R_t$, $m_t$ retains an exact 25% stationary marginal distribution over $\mathcal{M}$).

## V. Stage 2: Goal-Conditioned Bijective Field Permutation and Constant-Rate Chaff Entropy

Even if an agent runtime rotates wire serialization formats, encoding static field names or static numerical field IDs (such as Protobuf field tag `1` for `jsonrpc` and tag `2` for `method`) allows an L7 observer to correlate field positions across turns. Stage 2 eliminates structural schema invariants via a goal-conditioned 4-round Luby-Rackoff Feistel permutation network paired with a provably entropy-flattening constant-rate chaff engine.

### A. Balanced 4-Round Keyed Feistel Schema Network

Let $\mathcal{F}_{\text{canon}} = (f_1, f_2, \ldots, f_{64})$ denote the canonical registry of 64 standard MCP and A2A schema field names (including `jsonrpc`, `method`, `params`, `id`, `result`, `error`, `tool_name`, `arguments`, `session_id`, `agent_id`, `goal_hash`, etc.), assigned to integer slots $\tau(f_m) = m \in \{1, \ldots, 64\}$. Any unseen custom string key $f_{\text{cust}} \notin \mathcal{F}_{\text{canon}}$ encountered in a nested tool payload is dynamically assigned a collision-free session slot $\tau(f_{\text{cust}}) \in \{256, \ldots, 65533\}$ (supporting up to $|\mathcal{F}_{\text{custom}}(P_t)| \le 65{,}278$ distinct custom field names per turn), and the mapping $\{\tau(f_{\text{cust}}) \mapsto f_{\text{cust}}\}$ is stored inside a reserved symbol-table slot $\tau_{\text{sym}} = \texttt{0xFFFE} = 65534$ whose own slot identifier and internal keys are likewise permuted by $F_k$.

To permute the 16-bit slot domain $\mathbb{Z}_{2^{16}} = \{0, 1, \ldots, 65535\}$ at turn $t$, we derive four 256-byte round lookup tables (8-bit pseudorandom functions) $S_0^{(t)}, S_1^{(t)}, S_2^{(t)}, S_3^{(t)} : \{0, \ldots, 255\} \to \{0, \ldots, 255\}$ from a single invocation of the NIST FIPS 202 extendable-output function SHAKE-256 [9] seeded by $(R_t, H(\mathcal{G}_t))$:

$$S_0^{(t)} \parallel S_1^{(t)} \parallel S_2^{(t)} \parallel S_3^{(t)} = \text{SHAKE-256}\Big(\texttt{"chameleon-feistel-v1|"} \parallel R_t \parallel \texttt{"|"} \parallel H(\mathcal{G}_t),\ 1024\ \text{B}\Big). \tag{6}$$

Given any 16-bit slot integer $x \in \{0, \ldots, 65535\}$, we split $x$ into an 8-bit upper byte $L_0 = (x \gg 8)\ \&\ \texttt{0xFF}$ and an 8-bit lower byte $R_0 = x\ \&\ \texttt{0xFF}$. The forward 4-round Feistel map $F_k(x)$ iterates for $r = 0, 1, 2, 3$:

$$L_{r+1} = R_r, \qquad R_{r+1} = L_r \oplus S_r^{(t)}[R_r], \tag{7}$$

and outputs the 16-bit permuted tag $\tilde{\tau} = F_k(x) = (L_4 \ll 8) \mid R_4 \in \{0, \ldots, 65535\}$.

**Theorem 1** (Keyed Feistel Bijective Schema Invertibility and Luby-Rackoff PRP Security)**.** *Let $F_k : \{0, \ldots, 2^{16}-1\} \to \{0, \ldots, 2^{16}-1\}$ be the 4-round balanced Feistel network defined by Eqs. (6)–(7). Then:*

1) ***Exact Bijectivity and Closed-Form Inverse:*** *For every ratchet key $R_t \in \{0,1\}^{256}$ and every goal context $\mathcal{G}_t \in \{0,1\}^*$, $F_k$ is a strict bijection on $\{0, \ldots, 65535\}$. Its exact inverse $F_k^{-1} : \{0, \ldots, 65535\} \to \{0, \ldots, 65535\}$ is given by setting $(L_4, R_4) = ((y \gg 8)\ \&\ \texttt{0xFF},\ y\ \&\ \texttt{0xFF})$ and iterating in reverse for $r = 3, 2, 1, 0$:*

$$R_r = L_{r+1}, \qquad L_r = R_{r+1} \oplus S_r^{(t)}[L_{r+1}], \tag{8}$$

*returning $F_k^{-1}(y) = (L_0 \ll 8) \mid R_0 = x$. Consequently, recursive AST schema permutation and unpermutation satisfy $\text{Unpermute}_k(\text{Permute}_k(P_t)) = P_t$ with zero collisions and 100% structural fidelity.*

2) ***Strong Pseudorandom Permutation (sPRP) Bound:*** *Under the assumption that SHAKE-256 acts as a secure pseudorandom function family with distinguishing advantage $\epsilon_{PRF}$, the advantage of any adaptive chosen-plaintext/chosen-ciphertext adversary making $q_t$ field-tag queries within turn $t$ in distinguishing standalone $F_k$ from a uniformly sampled random permutation $\pi \xleftarrow{\$} \mathfrak{S}_{65536}$ is bounded by the Luby-Rackoff 4-round adaptive CCA bound [1], [25]:*

$$\text{Adv}_{F_k}^{sPRP}(\mathcal{A}) \le \frac{2q_t^2}{2^8} + \epsilon_{PRF}. \tag{9}$$

*Proof.* To establish Part (1), observe that for each individual round $r \in \{0, 1, 2, 3\}$, the round transformation $\psi_r(L_r, R_r) = (R_r,\ L_r \oplus S_r^{(t)}[R_r])$ is a map from $\mathbb{Z}_{256} \times \mathbb{Z}_{256}$ to itself. Defining the inverse map $\psi_r^{-1}(L_{r+1}, R_{r+1}) = (R_{r+1} \oplus S_r^{(t)}[L_{r+1}],\ L_{r+1})$ and substituting $L_{r+1} = R_r$ and $R_{r+1} = L_r \oplus S_r^{(t)}[R_r]$ yields:

$$\psi_r^{-1}(\psi_r(L_r, R_r)) = \Big((L_r \oplus S_r^{(t)}[R_r]) \oplus S_r^{(t)}[R_r],\ R_r\Big) = (L_r \oplus 0,\ R_r) = (L_r, R_r), \tag{10}$$

because bitwise XOR $\oplus$ over $\{0,1\}^8$ is an involution ($a \oplus b \oplus b = a$) regardless of whether the lookup table $S_r^{(t)}$ is itself injective. Since the composition of four bijections $F_k = \psi_3 \circ \psi_2 \circ \psi_1 \circ \psi_0$ is a bijection with inverse $F_k^{-1} = \psi_0^{-1} \circ \psi_1^{-1} \circ \psi_2^{-1} \circ \psi_3^{-1}$, no two distinct field slots $x_1 \neq x_2 \in \{0, \ldots, 65535\}$ can ever map to the same wire tag $F_k(x_1) = F_k(x_2)$. For recursive dictionary payloads $P_t$, canonical fields map to disjoint slots $\{1, \ldots, 64\}$ and custom fields map to disjoint slots $\{256, \ldots, 65533\}$ recorded in slot 65534; hence bijectivity of $F_k$ guarantees exact recursive invertibility.

To establish Part (2), by the classical Luby-Rackoff theorem [1] on 4-round balanced Feistel networks over $\{0,1\}^{2n}$ with half-block width $n = 8$ (see also Black and Rogaway [25] and Patarin [27] for small-domain Feistel analysis), replacing the ideal random round functions $f_0, \ldots, f_3 : \{0,1\}^8 \to \{0,1\}^8$ with the SHAKE-256-derived tables $S_0^{(t)}, \ldots, S_3^{(t)}$ incurs at most $\epsilon_{\text{PRF}}$ distinguishing advantage, and the transcript collision probability among $q_t$ adaptive forward/inverse queries in the internal 8-bit branches is bounded by $2\binom{q_t}{2}/2^8 + 2\binom{q_t}{2}/2^8 \leq 2q_t^2/2^8$. In the standalone unwhitened regime (Ablation Config C), Eq. (9) is non-vacuous for typical MCP tool schemas having $q_t \in [5, 11] < 16$ fields per turn, and because Agent-Chameleon re-keys $(S_0^{(t)}, S_1^{(t)}, S_2^{(t)}, S_3^{(t)})$ on every RPC turn $t$ via $(R_t, H(\mathcal{G}_t))$, an adversary cannot accumulate queries across turns against a fixed permutation. Furthermore, in the full Agent-Chameleon pipeline, all serialized tags $F_k(\tau_i)$ are subsequently XOR-whitened by the Stage 2B SHAKE-256 stream $K_{\text{stream}}$, preventing a wire adversary from observing raw Feistel outputs even when $q_t \geq 16$. □

### *B. Set-Complement and Affine-Bijection Constant-Rate Chaff Engine*

While polymorphic serialization and Feistel field permutation eliminate plaintext AST field names, the raw serialized byte string $S_t \in \{0, \ldots, 255\}^{L_s}$ still exhibits two statistical side channels: (i) its byte length $L_s$ correlates with the number and size of parameters in the invoked tool schema, and (ii) even if $S_t$ is XOR-whitened by a stream cipher of finite length $L_s \in [100, 500]$ bytes, the plug-in Shannon byte entropy of an i.i.d. uniform byte string of length $N$ is downward-biased by finite-sample birthday collisions ($\mathbb{E}[H_N] \approx 8 - \frac{255}{2N \ln 2} \approx 7.64$ bits/byte at $N = 512$), and naive padding (such as zero-padding or deterministic whitespace) creates sharp histogram spikes that traffic classifiers readily exploit [18].

To achieve both constant-tier length quantization and single-frame plug-in Shannon byte entropy $H(B_t) \geq 7.980$ bits/byte in under $40\,\mu$s of CPU execution, Stage 2B encapsulates $S_t$ as follows:

1) **Quantum Bucket Selection:** Given payload length $L_s = |S_t|$ and an 8-byte inner framing header $H_{\text{in}} = \text{uint32_be}(\texttt{0xC4A3E100} \mid m_t) \parallel \text{uint32_be}(L_s)$ (total prefix length $L_p = L_s + 8$), the engine selects the smallest discrete bucket tier $B \in \mathcal{B} = \{512, 1024, 2048, 3072, 4096, 6144, 8192\}$ satisfying $B \geq \max(B_{\min}, 4L_p, 1024)$ (or $B = 512$ when $B_{\min} = 512$ and $L_p \leq 64$, i.e., $L_s \leq 56$ B), yielding a chaff budget $L_c = B - L_p \geq 3L_p$.
2) **SHAKE-256 Stream Whitening:** The engine expands a per-turn keystream $K_{\text{stream}} = \text{SHAKE-256}(\texttt{"chameleon-stream-v1|"} \parallel R_t \parallel \text{uint64_be}(t),\ L_p + 64)$ and XOR-whitens the raw prefix in a single C-level big-integer operation:
$$W_p = (H_{\text{in}} \parallel S_t) \oplus K_{\text{stream}}[0 : L_p]. \tag{11}$$
3) **Slice-Wise Set-Complement Histogram Balancing:** Partition $W_p$ into $S = \lceil L_p/128 \rceil$ contiguous slices $W_p^{(0)}, \ldots, W_p^{(S-1)}$ of length $L_s^{(s)} \leq 128$ bytes. For each slice $W_p^{(s)}$, let $\text{set}(W_p^{(s)}) \subset \Sigma_{256} = \{0, \ldots, 255\}$ denote the set of distinct byte values occurring in $W_p^{(s)}$, with intra-slice duplicate count $d_s = L_s^{(s)} - |\text{set}(W_p^{(s)})|$. The engine computes the exact set complement $M^{(s)} = \Sigma_{256} \setminus \text{set}(W_p^{(s)})$ of length $256 - |\text{set}(W_p^{(s)})|$ (containing every byte value absent from $W_p^{(s)}$), circularly shifts $M^{(s)}$ by the pseudorandom offset $K_{\text{stream}}[L_p + s] \bmod |M^{(s)}|$, and appends $M^{(s)}$ to the chaff buffer. Thus each combined slice $U^{(s)} = W_p^{(s)} \parallel M^{(s)}$ has length $256 + d_s$ and guarantees that every byte $b \in \Sigma_{256}$ appears at least once.
4) **Keyed $\mathbb{Z}_{256}$ Affine-Bijection Block Filling:** The remaining chaff capacity after the $S$ complemented slices is $L_{\text{rem}} = B - \sum_{s=0}^{S-1}(256 + d_s) = 256(M - S) - C_{\text{dup}}$, where $M = B/256$ and $C_{\text{dup}} = \sum_{s=0}^{S-1} d_s$ is the total duplicate count across $W_p$. Writing $L_{\text{rem}} = 256q + r_{\text{rem}}$ with $q = \lfloor L_{\text{rem}}/256 \rfloor$ and $r_{\text{rem}} = L_{\text{rem}} \bmod 256$, the engine appends $q$ complete 256-byte affine permutation blocks (plus a final prefix slice of length $r_{\text{rem}}$ when $r_{\text{rem}} > 0$) drawn from the precomputed family of 128 multiplicative-odd bijections over $\mathbb{Z}_{256}$:
$$\pi_{a,r}(x) = ((2a+1) \cdot ((x + r) \bmod 256)) \bmod 256, \quad a \in \{0, \ldots, 127\}, \quad r \in \{0, \ldots, 255\}, \tag{12}$$
where $(a, r)$ are indexed by consecutive bytes of $K_{\text{stream}}$. Because $\gcd(2a + 1, 256) = 1$, every complete 256-byte block $\Pi_{a,r} = (\pi_{a,r}(0), \ldots, \pi_{a,r}(255))$ is a strict bijection on $\{0, \ldots, 255\}$ containing every byte value with exact multiplicity 1.

**Theorem 2** (Plug-In Shannon Byte-Entropy Lower Bound and Histogram Uniformity)**.** *Let $B_t \in \{0, \ldots, 255\}^B$ be a wire frame constructed by the Stage 2B Set-Complement and Affine-Bijection Chaff Engine with prefix length $L_p \leq B/4$ and bucket size $B = 256M$ ($M \geq 4$). Let $c(b) = |\{i \in \{0, \ldots, B-1\} : B_t[i] = b\}|$ denote the empirical frequency count of byte $b \in \{0, \ldots, 255\}$ in $B_t$, and let $p(b) = c(b)/B$. Under the operational invariant $C_{dup} \leq 256$ (which holds with probability $> 1 - 10^{-18}$ for SHAKE-256 whitened prefixes of length $L_p \leq B/4$, where $\mathbb{E}[d_s] \leq 29.5$ per 128-byte slice):*

1) ***Guaranteed Full Alphabet Support and Bounded Multiplicity Variance:*** *Every byte value $b \in \{0, \ldots, 255\}$ occurs at least $c(b) \geq S + q = M - \lceil C_{dup}/256 \rceil \geq$*

$M-1 \geq 3$ *times in* $B_t$*, and the expected squared histogram deviation from the uniform distribution* $u(b) = 1/256$ *(accounting for both intra-slice collisions and the truncated affine tail of length* $r_{rem} \in [0, 255]$*) satisfies:*

$$\mathbb{E}\left[\sum_{b=0}^{255}\left(p(b)-\frac{1}{256}\right)^2\right] = \frac{1}{B^2}\mathbb{E}\left[\sum_{b=0}^{255}(c(b)-M)^2\right] \leq \frac{L_p+64}{B^2} \leq \frac{M+1}{1024M^2}. \tag{13}$$

2) ***Single-Frame Shannon Byte-Entropy Lower Bound:*** *Defining* $\kappa_M = \frac{M}{2(M-1)} \leq \frac{2}{3}$ *for* $M \geq 4$*, the empirical plug-in Shannon byte entropy* $H(B_t) = -\sum_{b=0}^{255} p(b)\log_2 p(b)$ *satisfies the exact deterministic lower bound:*

$$H(B_t) \geq 8.000 - \frac{256\kappa_M}{\ln 2}\sum_{b=0}^{255}\left(p(b)-\frac{1}{256}\right)^2,$$
$$\mathbb{E}[H(B_t)] \geq 8.000 - \frac{0.2404(M+1)}{M^2} \textit{ bits/B}, \tag{14}$$

*which yields a worst-case deterministic floor* $H(B_t) \geq 7.955$ *bits/byte at* $M = 4$*, an expected entropy* $\mathbb{E}[H(B_t)] \geq 7.990$ *bits/byte for typical payloads (*$L_p \approx 250$ *B in* $B = 1024$ *B where* $\mathbb{E}[\sum(c(b)-M)^2] \approx 72$*), and a high-probability concentration floor* $H(B_t) \geq 7.980$ *bits/byte across* $100\%$ *of operational traces (*$7.992 \pm 0.005$ *bits/byte empirical mean over* $N = 1{,}000$ *traces).*

*Proof.* Each combined slice $U^{(s)} = W_p^{(s)} \parallel M^{(s)}$ contributes $+1$ to all 256 byte bins plus $d_s = L_s^{(s)} - |\operatorname{set}(W_p^{(s)})|$ duplicate increments, and each of the $q = M - S - \lceil C_{\text{dup}}/256 \rceil$ complete 256-byte affine blocks $\Pi_{a,r}$ contributes exactly $+1$ to every bin $b \in \{0, \ldots, 255\}$ because $\gcd(2a+1, 256) = 1$ makes $x \mapsto (2a+1)(x+r) \bmod 256$ a unit-multiplier bijection on $\mathbb{Z}_{256}$. When $0 < C_{\text{dup}} \leq 256$, we have $q = M - S - 1$ and the final partial affine slice of length $r_{\text{rem}} = 256 - C_{\text{dup}}$ contributes $+1$ to $r_{\text{rem}}$ distinct bins and 0 to the remaining $256 - r_{\text{rem}} = C_{\text{dup}}$ bins; hence every bin satisfies $c(b) \geq S + q = M - 1 \geq 3$.

Writing $\delta_b = p(b) - \frac{1}{256} = (c(b) - M)/B$ and $x_b = 256\delta_b = \frac{c(b)-M}{M} \geq -\frac{1}{M}$, we have $\sum_{b=0}^{255} x_b = 0$. By Taylor's theorem with exact Lagrange remainder on $g(x) = (1+x)\ln(1+x)$ for $x \geq -1/M$, since $g(0) = 0$, $g'(0) = 1$, and $g''(\xi) = \frac{1}{1+\xi} \leq \frac{1}{1-1/M} = \frac{M}{M-1} = 2\kappa_M$ for all $\xi \in [-1/M, x_b]$, we obtain the exact upper bound $(1+x_b)\ln(1+x_b) \leq x_b + \kappa_M x_b^2$ without dropping higher-order terms. Substituting $p(b) = \frac{1+x_b}{256}$ yields:

$$H(B_t) = 8 - \frac{1}{256\ln 2}\sum_{b=0}^{255}(1+x_b)\ln(1+x_b) \geq 8 - \frac{256\kappa_M}{\ln 2}\sum_{b=0}^{255}\delta_b^2. \tag{15}$$

Under SHAKE-256 whitening, the duplicate excess across the $S$ slices has second moment $\leq L_p$, while the independent partial affine tail of $r_{\text{rem}}$ without-replacement draws adds variance $r_{\text{rem}}(1 - r_{\text{rem}}/256) \leq 64$, giving $\mathbb{E}[\sum_{b=0}^{255}(c(b)-M)^2] \leq L_p + 64 \leq 64(M+1)$ and Eq. (14). □

## VI. Stage 3: Sub-35-Microsecond Schnorr Zero-Knowledge Challenge-Response

To authenticate each morphed frame $B_t$ without transmitting static API keys, reusable OAuth bearer tokens, or slow per-message public-key signatures, Stage 3 implements a Dual-Tier Zero-Knowledge Session Handshake carried in the `X-Chameleon-Proof-Nonce` wire header. Crucially, pairing an algebraic Schnorr Non-Interactive Zero-Knowledge Proof of Knowledge (NIZKPoK) $(R_{\text{com}}, e_t, s_t, Y_t)$ with a 128-bit symmetric ratchet authenticator $\tau_{\text{ratchet}}$ serves two distinct verification planes: (i) zero-trust L7 service-mesh gateways and ingress middleboxes that hold only the turn's public commitment $Y_t$ (without possessing the secret session ratchet key $R_t$ needed to decrypt $B_t$) can algebraically verify $g^{s_t} \equiv R_{\text{com}} Y_t^{e_t} \pmod p$ at the network edge to drop unauthenticated packets before they reach the agent runtime, while (ii) the receiving agent endpoint holding $R_t$ simultaneously verifies $\tau_{\text{ratchet}}$ and $Y_t = g^{x_t(R_t)} \bmod p$ for 128-bit symmetric unforgeability.

### A. Protocol Specification and Dual-Tier Field Arithmetic

Let $p$ be a prime modulus and $g \in \mathbb{F}_p^*$ be a generator of a multiplicative subgroup of order $q_{\text{ord}} = \operatorname{ord}_p(g)$. Agent-Chameleon supports two cryptographic operating modes selectable per deployment policy:

- **Tier-1 Hot-Path Mode (`fast_m127`):** Operates over the Mersenne prime field $p = M_{127} = 2^{127} - 1$ with generator $g = 7$, using single-turn ephemeral witnesses of byte length $\ell_x = 4$ B ($x_t \in [1, 2^{32} - 1]$), blinding nonces of byte length $\ell_k = 5$ B ($k_t \in [1, 2^{40} - 1]$), and Fiat-Shamir challenges of byte length $\ell_e = 4$ B ($e_t \in [1, 2^{32} - 1]$), refreshed on every RPC turn via 384-bit BLAKE2b expansion of $(R_t, t)$ and paired with the 128-bit BLAKE2s ratchet tag $\tau_{\text{ratchet}}$. Because each witness $x_t$ is used for exactly one turn ($Q = 1$) and guarded by the 128-bit MAC $\tau_{\text{ratchet}}$ against online forgery, 32–40 bit exponents suffice on the hot path while requiring only 32–40 modular squarings in C, completing proof generation in $20.92\,\mu$s mean ($20.38\,\mu$s p50) and verification in $28.64\,\mu$s mean ($28.58\,\mu$s p50). (When standalone statistical zero-knowledge without subgroup reduction is required against an offline transcript analyzer, setting $\ell_k = 16$ B provides $2^{-64}$ statistical blinding at $+6.1\,\mu$s, or computing $s_t = (k_t + e_t x_t) \bmod q_{\text{ord}}$ provides perfect HVZK.)
- **Tier-2 Full-256-Bit & Ed25519 Anchor Mode (`full256 + ed25519`):** Operates over the 255-bit prime field $p = 2^{255} - 19$ with generator $g = 5$, witness length $\ell_x = 24$ B (192 bits), blinding nonce length $\ell_k = 32$ B (256 bits), and challenge length $\ell_e = 16$ B (128 bits), alongside periodic RFC 8032 Ed25519 [7] signatures over the epoch ratchet root $R_t$ (transmitted in the optional `X-Chameleon-Epoch-Anchor` header every $\Delta_{\text{anchor}} = 100$ turns) for third-party cross-domain non-repudiation.

At turn $t$, both endpoints deterministically derive the turn's public session commitment $Y_t = g^{x_t} \bmod p$ from their synchronized ratchet key $R_t$ via:

$$\begin{aligned} d_t &= \text{BLAKE2b-384}\big(\texttt{"chameleon-zk-witness|"} \\ &\quad \| R_t \| \text{uint64_be}(t)\big), \\ x_t &= \text{int}(d_t[0:\ell_x]) \mid 1, \quad k_t = \text{int}(d_t[\ell_x : \ell_x + \ell_k]) \mid 1, \\ Y_t &= g^{x_t} \bmod p. \end{aligned} \tag{16}$$

To prove knowledge of $x_t$ bound to the wire frame $B_t$ without revealing $x_t$ or $R_t$, the sender computes the Schnorr commitment $R_{\text{com}}$, Fiat-Shamir challenge $e_t$ [2], [3], algebraic response $s_t$, and 128-bit ratchet tag $\tau_{\text{ratchet}}$:

$$\begin{aligned} R_{\text{com}} &= g^{k_t} \bmod p, \qquad h_B = \text{BLAKE2s-128}(B_t), \\ e_t &= \text{int}\Big(\text{BLAKE2s}\big(\texttt{"chameleon-fs|"} \| R_{\text{com}} \\ &\quad \| Y_t \| \text{uint64_be}(t) \| h_B\big)[0:\ell_e]\Big) \mid 1, \\ s_t &= k_t + e_t \cdot x_t \quad (\text{in } \mathbb{Z}, \text{ or } \bmod q_{\text{ord}}), \\ \tau_{\text{ratchet}} &= \text{BLAKE2s-128}\Big(\texttt{"chameleon-ratchet-tag|"} \\ &\quad \| R_t \| \text{uint64_be}(t) \| h_B \| R_{\text{com}}\Big). \end{aligned} \tag{17}$$

The tuple $\pi_t = (t, R_{\text{com}}, e_t, s_t, Y_t, \tau_{\text{ratchet}})$ is packed into a compact 81-byte binary structure in `fast_m127` mode (108 ASCII characters in URL-safe Base64) or a 153-byte structure in `full256` mode (204 Base64 characters) and transmitted in the `X-Chameleon-Proof-Nonce` header.

**Theorem 3** (Non-Interactive Schnorr Ratchet Soundness, Zero-Knowledge Simulatability, and Replay Immunity)**.** *The Stage 3 handshake protocol defined by Eqs.* (16)–(17) *satisfies:*

1) ***Perfect Completeness:*** *For any honest prover and verifier holding synchronized ratchet state $R_t$ and receiving unmodified frame $B_t$, the algebraic verification identity*
$$g^{s_t} \equiv R_{com} \cdot Y_t^{e_t} \pmod p \tag{18}$$
*and tag equality $\tau_{ratchet} = \hat{\tau}_{ratchet}$ hold with probability* 1.
2) ***Special Soundness and 128-Bit Unforgeability:*** *Under the Discrete Logarithm assumption in $\langle g \rangle \subset \mathbb{F}_p^*$ of order $q_{ord} = \text{ord}_p(g)$ and the Random Oracle / PRF security of* BLAKE2s *[29], [30], any adversary who produces two accepting transcripts $(R_{com}, e_1, s_1)$ and $(R_{com}, e_2, s_2)$ with $e_1 \neq e_2$ immediately yields the secret witness $x_t \equiv (s_1 - s_2)(e_1 - e_2)^{-1} \pmod{q_{ord}}$ (or $x_t = (s_1 - s_2)/(e_1 - e_2)$ over $\mathbb{Z}$), and the probability that an unauthenticated adversary without $R_t$ forges a valid $(\pi_t, B_t)$ at turn $t$ is at most $2^{-128}$.*
3) ***Honest-Verifier / Random-Oracle Zero-Knowledge:*** *Under modular response $s_t = (k_t + e_t x_t) \bmod q_{ord}$ (or over $\mathbb{Z}$ with statistical blinding $k_t \gg e_t x_t$), the public transcript $(R_{com}, e_t, s_t, Y_t)$ can be simulated in the Random Oracle Model without knowing $x_t$, and $x_t$ is permanently retired after single-turn use at epoch $t$.*
4) ***Replay and Desynchronization Immunity:*** *Any replayed packet $W_{t'} (t' \leq t)$ or tampered packet $\tilde{W}_t$ is rejected with probability $1 - 2^{-128}$, and atomic ratchet rollback leaves the verifier's state $(R_{t-1}, t, D_{t-1})$ intact for the legitimate turn-$t$ packet.*

*Proof.* For Part (1), since $s_t = k_t + e_t x_t$ holds over $\mathbb{Z}$ (or modulo $q_{\text{ord}} = \text{ord}_p(g)$), standard exponent laws in $(\mathbb{F}_p^*, \cdot)$ immediately yield:

$$\begin{aligned} g^{s_t} \bmod p &= g^{k_t + e_t x_t} \bmod p = \big(g^{k_t} \cdot (g^{x_t})^{e_t}\big) \bmod p \\ &= (R_{\text{com}} \cdot Y_t^{e_t}) \bmod p. \end{aligned} \tag{19}$$

For Part (2), by the Pointcheval-Stern Forking Lemma [30], rewinding a successful Schnorr forger on the random oracle $\text{BLAKE2s}(\texttt{"chameleon-fs|"} \| \dots)$ yields two transcripts sharing $R_{\text{com}} = g^{k_t} \bmod p$ with distinct challenges $e_1 \neq e_2$ satisfying $g^{s_1} \equiv R_{\text{com}} Y_t^{e_1}$ and $g^{s_2} \equiv R_{\text{com}} Y_t^{e_2} \pmod p$, whence $g^{s_1 - s_2} \equiv Y_t^{e_1 - e_2} = g^{x_t(e_1 - e_2)} \pmod p$. Consequently, $s_1 - s_2 \equiv x_t(e_1 - e_2) \pmod{q_{\text{ord}}}$, extracting $x_t \bmod q_{\text{ord}}$. Moreover, because the endpoint verifier simultaneously checks the 128-bit keyed MAC $\tau_{\text{ratchet}} = \text{BLAKE2s-128}(R_t \| t \| h_B \| R_{\text{com}})$ and verifies $Y_t == g^{x_t(R_t)} \bmod p$, any external attacker lacking $R_t$ faces a $2^{-128}$ forgery barrier on $\tau_{\text{ratchet}}$ even before the algebraic Schnorr check is reached.

For Part (4), the verifier maintains a monotonic epoch counter $t_{\text{exp}}$ and a sliding anti-replay set $\mathcal{W}_{\text{seen}}$ of $(t, \tau_{\text{ratchet}})$ tuples over window $W = 4096$. Any duplicate $(t, \tau_{\text{ratchet}}) \in \mathcal{W}_{\text{seen}}$ or mismatched header epoch $t \neq t_{\text{exp}}$ is rejected in $\mathcal{O}(1)$ hash-table lookup time. Prior to speculative ratchet advancement, the verifier checkpoints $(R_{\text{saved}}, t_{\text{saved}}, D_{\text{saved}}) \leftarrow (R_{t-1}, t, D_{t-1})$ and restores the checkpoint inside the `except` handler whenever proof verification, chaff unwrapping, or MAC validation fails, ensuring that an active attacker flooding invalid packets cannot advance or corrupt the receiver's ratchet state. □

## VII. Implementation and Wire-Protocol Specification

We implemented Agent-Chameleon in Python 3.11 (`agent_chameleon/`) using standard C-accelerated cryptographic primitives (`hashlib.shake_256`, `hashlib.blake2s`, `hashlib.blake2b`, `hmac.digest`, and `cryptography.hazmat.primitives.asymmetric` for X25519 and Ed25519).

### A. Wire Header Contract and Packet Framing

Every outbound RPC frame sealed by `ChameleonMorphEngine.seal_message` emits four deterministic HTTP/2, gRPC, or FastMCP transport metadata headers:

1) `Content-Type`: set to `application/x-chameleon-morph+octet-stream` — signals to the receiving Agent-Chameleon sidecar or endpoint middleware that the payload body is encapsulated in a constant-rate morphed envelope.
2) `X-Chameleon-Epoch-ID`: `<16-hex-digits>` — carries the big-endian 64-bit monotonic turn index $t$ formatted as a 16-character lowercase hexadecimal string (e.g., `000000000000002a` for $t = 42$).

3) `X-Chameleon-Proof-Nonce`: `<base64url-token>` — carries the serialized `SchnorrProof` structure $\pi_t$ (81 raw bytes / 108 URL-safe Base64 characters in `fast_m127` mode, or 153 raw bytes / 204 Base64 characters in `full256` mode).
4) `X-Chameleon-Bucket-Tier`: `<integer-bytes>` — declares the discrete quantum bucket size $B \in \{512, 1024, 2048, 3072, 4096, 6144, 8192\}$ of the wire body $B_t$ for zero-copy pre-allocation on the receiver socket buffer.

Inside the $B$-byte wire body $B_t$, the first 8 bytes (prior to SHAKE-256 stream whitening) form the inner framing header:

$$H_{\text{in}} = \text{uint32_be}\left(\texttt{0xC4A3E100} \mid (m_t \,\&\, \texttt{0xFF})\right) \\ \| \ \text{uint32_be}(L_s), \quad m_t \in \{0..3\},\ L_s = |S_t|. \tag{20}$$

When the receiver invokes `unwrap_chaff` on `ChaffEntropyEngine`, it generates the first 8 bytes of the SHAKE-256 keystream, XOR-unwhitens $H_{\text{in}}$, verifies that the upper 24 bits equal the magic constant `0xC4A3E1`, extracts the polymorphic format code $m_t \in \{0, 1, 2, 3\}$ and body length $L_s$, validates the bounds guard $0 \le L_s \le B - 8$ (raising a `ValueError` that triggers atomic ratchet rollback if violated), and slices out the payload $W_p[8 : 8 + L_s]$ while discarding the trailing $B - (8 + L_s)$ chaff bytes in $\mathcal{O}(1)$ pointer arithmetic without having to invert the chaff blocks.

### B. Algorithmic Workflow of Sender and Receiver Pipelines

Table I summarizes the byte-level structure of both the `X-Chameleon-Proof-Nonce` header (for both `fast_m127` and `full256` modes) and the $B$-byte morphed wire body $B_t$. On the sender side, `seal_message` executes five deterministic steps:

1) **Step S1 (Ratchet Advance):** Compute $H(\mathcal{G}_t)$, derive $(t, R_t, m_t) \leftarrow \texttt{ratchet.advance}(\mathcal{G}_t)$ via Eq. (4) and Eq. (5).
2) **Step S2 (Feistel Schema Permutation):** Instantiate $\texttt{FeistelSchemaPermuter}(R_t, \mathcal{G}_t)$, expand the four 256-byte round S-boxes $(S_0^{(t)}, S_1^{(t)}, S_2^{(t)}, S_3^{(t)})$ via a single 1024-byte SHAKE-256 call, and recursively map all string keys in $P_t$ to 16-bit permuted integer tags $\tilde{\tau}_i = F_k(\tau_i)$ (storing any unseen custom keys in permuted symbol slot $F_k(\texttt{0xFFFE})$).
3) **Step S3 (Polymorphic Serialization):** Dispatch the permuted dictionary $\tilde{P}_t$ to the turn's selected wire encoder ($m_t = 0$: `encode_protobuf_varint`; $m_t = 1$: `encode_flatbuffers_table`; $m_t = 2$: `encode_cbor_rfc8949`; $m_t = 3$: `encode_canonical_json_rpc`) to obtain $S_t \in \{0, \ldots, 255\}^{L_s}$.
4) **Step S4 (Constant-Rate Chaff & Stream Whitening):** Prepend the 8-byte header $H_{\text{in}}$, XOR-whiten $H_{\text{in}} \ \| \ S_t$ with SHAKE-256 using C-level big-integer XOR (`int.from_bytes / to_bytes` in `_xor_bytes_fast`), append the circularly rotated set-complement $\Sigma_{256} \setminus \text{set}(W_p^{(s)})$ for each 128-byte prefix slice, and pad the remainder to bucket size $B$ using precomputed $\mathbb{Z}_{256}$ affine bijection blocks $\Pi_{a,r}$ to obtain $B_t$.
5) **Step S5 (Schnorr NIZKPoK & Transcript Commit):** Generate $\pi_t \leftarrow \texttt{zk_engine.generate_proof}(R_t, t, B_t)$, update the rolling frame digest $D_t \leftarrow \text{SHA256}(B_t)$, and emit $\texttt{ChameleonWirePacket}(t, m_t, \mathcal{H}_t, B_t)$.

## VIII. Empirical Evaluation

We evaluate Agent-Chameleon across $N = 1{,}000$ multi-agent tool invocation traces spanning $|\mathcal{C}| = 20$ distinct enterprise MCP and A2A tool schemas (listed in Table VI), plus a multi-tier payload scaling suite spanning bucket quanta $B \in \{512, 1024, 2048, 4096, 8192\}$ bytes. Single-node latency and classifier experiments were executed on a 24-core Apple Silicon workstation (Python 3.11) with deterministic random seeds (`seed=20261104`), distributed swarm throughput scaling ($1 \rightarrow 128$ concurrent agents) was evaluated across a 6-node cluster (144 physical CPU cores total), and all invariants were validated by our 24-test automated verification suite (`tests/test_agent_chameleon.py`).

### A. Experimental Methodology and Baselines

To evaluate both cryptographic overhead and statistical traffic-analysis resistance, we compare six protocol configurations across all $N = 1{,}000$ traces (50 invocations per tool class):

1) **Baseline 1 — Static MCP/A2A JSON-RPC:** Standard unpadded JSON-RPC 2.0 serialization with plaintext schema keys (`"jsonrpc"`, `"method"`, `"params"`, `"tool_name"`, `"arguments"`).
2) **Baseline 2 — Length-Padded JSON-RPC (512B Quantum):** Standard JSON-RPC 2.0 zero-padded to the next multiple of 512 bytes to emulate transport-layer record length padding [10], [17] without L7 schema permutation or byte whitening.
3) **Baseline 3 — Static Binary CBOR (RFC 8949):** Binary RFC 8949 CBOR encoding [5] without constant-rate chaff or stream whitening.
4) **Baseline 4 — Standard Stream Whitening + 1024B Random Padding:** Encrypts each payload with SHAKE-256 stream whitening and pads to $B = 1024$ bytes using i.i.d. pseudorandom bytes (emulating standard TLS 1.3 AEAD record padding without set-complement or affine-bijection histogram balancing).
5) **Ablation — Polymorphic + Feistel Only (No Chaff):** Stage 1 4-way polymorphic codec rotation plus Stage 2A 4-round Feistel field permutation, omitting Stage 2B constant-rate chaff and stream whitening.
6) **Agent-Chameleon (Full 3-Stage Pipeline):** Complete Stage 1 + Stage 2 + Stage 3 pipeline with cycle-balanced 4-format polymorphism, 4-round SHAKE-256 Feistel permutation, set-complement plus affine-bijection

TABLE I
BYTE-LEVEL STRUCTURAL LAYOUT OF THE AGENT-CHAMELEON WIRE PACKET ($W_t = (\mathcal{H}_t, B_t)$) ACROSS `FAST_M127` (81 B) AND `FULL256` (153 B) MODES

| Envelope Layer | Field / Segment Name | Offset (`m127` / `full256`) | Size (`m127` / `full256`) | Cryptographic Construction & Functional Role |
|---|---|---|---|---|
| **Header Token** (`X-Chameleon-Proof-Nonce`, 81 B / 153 B raw, 108 / 204 B64) | `mode_byte` | 0 / 0 | 1 B / 1 B | `0x01` (`fast_m127` hot-path) or `0x02` (`full256` anchor mode) |
| | `epoch_id` ($t$) | 1 / 1 | 8 B / 8 B | Big-endian 64-bit monotonic turn counter uint64_be($t$) |
| | `commitment_r` ($R_{\text{com}}$) | 9 / 9 | 16 B / 32 B | Schnorr blinding commitment $R_{\text{com}} = g^{k_t} \bmod p$ |
| | `challenge_e` ($e_t$) | 25 / 41 | 8 B / 16 B | Fiat-Shamir challenge BLAKE2s($R_{\text{com}} \parallel Y_t \parallel t \parallel h_B$)[$0 : \ell_e$] |
| | `response_s` ($s_t$) | 33 / 57 | 16 B / 48 B | Algebraic Schnorr response $s_t = k_t + e_t x_t$ |
| | `public_y` ($Y_t$) | 49 / 105 | 16 B / 32 B | Per-turn ephemeral public key $Y_t = g^{x_t} \bmod p$ |
| | `ratchet_tag` ($\tau_{\text{ratchet}}$) | 65 / 137 | 16 B / 16 B | 128-bit BLAKE2s MAC over $(R_t, t, h_B, R_{\text{com}})$ |
| **Morphed Body** ($B_t \in \{0..255\}^B$, $B \in \{512..8192\}$) | Whitened Inner Header ($\tilde{H}_{\text{in}}$) | 0 | 8 B | (uint32_be(`0xC4A3E100` $\mid m_t$) $\parallel$ uint32_be($L_s$)) $\oplus$ $K_{\text{stream}}[0:8]$ |
| | Whitened Polymorphic Payload ($\tilde{S}_t$) | 8 | $L_s$ B | $S_t \oplus K_{\text{stream}}[8:8+L_s]$, where $S_t = \text{Encode}_{m_t}(\text{Permute}_{F_k}(P_t))$ |
| | Set-Complement Chaff Slices ($M^{(s)}$) | $8 + L_s$ | $\sum_s \lvert M^{(s)} \rvert$ | Circularly shifted missing alphabet $\Sigma_{256} \setminus \text{set}(W_p^{(s)})$ per 128 B slice |
| | Affine-Bijection Chaff Tail ($\Pi_{a,r}$) | $8 + L_s + \sum_s \lvert M^{(s)} \rvert$ | $L_{\text{rem}}$ B | Keyed 256 B bijections $\pi_{a,r}(x) = (2a+1)(x+r) \bmod 256$ padding to $B$ |

chaff padding, and `fast_m127` Fiat-Shamir Schnorr NIZKPoK authentication.

For adversarial traffic fingerprinting, we extract a 36-dimensional feature vector $\phi(W_t) \in \mathbb{R}^{36}$ from every wire frame (normalized frame length residual, plug-in Shannon byte entropy $H(B_t)/8$, ASCII printable character ratio, JSON structural delimiter density, and a 32-bin byte-value histogram; see Appendix F) and evaluate three distinct machine-learning classifiers trained on an $80\%/20\%$ stratified split (800 training traces, 200 held-out test traces across the 20 tool classes, where uniform random guessing yields $1/20 = 5.00\%$ accuracy): (i) a 50-estimator Random-Subspace Nearest-Centroid Ensemble (**Random Forest / Subspace Ensemble**), (ii) a 2-layer Multi-Layer Perceptron (**MLP**, $36 \to 48 \to 20$ with ReLU activations and cross-entropy SGD), and (iii) a distance-weighted 5-Nearest Neighbor ($k$-**NN**) classifier.

### B. Shannon Byte Entropy and Classifier Evasion Results

Figure 2 and Table II present the empirical Shannon byte entropy and attacker classification accuracy across all protocol configurations. On unprotected Static MCP/A2A JSON-RPC, the high redundancy of ASCII field names and JSON delimiters restricts plug-in Shannon byte entropy to a mean of $4.788$ bits/byte (p50: $4.773$, p95: $4.948$). Consequently, the attacker's Random-Subspace Ensemble, MLP, and $k$-NN classifiers achieve $95.0\%$, $94.5\%$, and $100.0\%$ accuracy ($96.50\%$ ensemble mean) in identifying the exact enterprise tool invoked.

Crucially, Table II confirms Dyer et al.'s "Peek-a-Boo" finding [18] in the multi-agent domain: merely zero-padding JSON-RPC frames to 512-byte length multiples (**Length-Padded JSON-RPC**) lowers plug-in byte entropy to $3.142$ bits/byte (due to the zero-byte spike) and barely reduces attacker accuracy ($89.5\%$ RF, $88.5\%$ MLP, $100.0\%$ $k$-NN; $92.67\%$ ensemble mean), because L7 non-zero byte-value histograms and delimiter densities remain intact. Enabling Stage 1 polymorphic rotation and Stage 2A Feistel field permutation without chaff (**Polymorphic + Feistel, No Chaff**) raises mean entropy to $5.538$ bits/byte and cuts RF accuracy from $95.0\%$ to $26.0\%$ ($36.17\%$ ensemble mean), as static field keys disappear and each turn cycles across four binary/text codecs. However, payload lengths and unwhitened value distributions still leak partial schema identity ($26.0\%$–$47.0\%$, well above the $5.0\%$ random baseline). Even when standard stream whitening is paired with 1024-byte i.i.d. pseudorandom padding (**Standard Whitening + 1024B Random Padding**), finite-sample birthday collisions across $N = 1024$ bytes cap empirical plug-in Shannon entropy at $7.752 \pm 0.018$ bits/byte (strictly below the $7.980$ bits/byte G2 threshold, since $\mathbb{E}[H_{\text{iid}}(1024)] \approx 8 - \frac{255}{2048 \ln 2} = 7.820$ bits/byte).

When the full **Agent-Chameleon** pipeline is active, the Stage 2B set-complement and affine-bijection chaff engine eliminates birthday-collision histogram variance and drives single-frame plug-in Shannon byte entropy to a mean of $7.992 \pm 0.005$ bits/byte (p50: $7.994$, p95: $7.996$, p99: $7.996$), within $0.008$ bits/byte of the theoretical $8.000$ bits/byte ceiling. Across the held-out test set, attacker accuracy collapses to $5.00\%$ for Random-Subspace Ensemble, $3.00\%$ for MLP, and $6.00\%$ for $k$-NN, yielding an ensemble mean of $4.67\%$—statistically indistinguishable from the $1/20 = 5.00\%$ uniform random-guess baseline (representing a $20.7\times$ reduction in attacker classification rate). Furthermore, in our multi-tier payload scaling experiment where tool arguments span $L_s \in [40, 1800]$ bytes across bucket tiers $B \in \{512, 1024, 2048, 4096, 8192\}$, classifier accuracy within each bucket tier remains at $4.80\% \pm 0.45\%$ ($H(B_t) \in [7.984, 7.998]$ bits/byte), and optional keyed Fisher-Yates 64-byte block interleaving ensures that even sub-slice positional histograms $B_t[a:b]$ exhibit identical $7.99$ bits/byte uniformity.

### C. Microsecond Latency Breakdown and Swarm Throughput

Figure 3 and Table III report the microsecond execution latency of each Agent-Chameleon subsystem across $N = 1{,}000$ traces, alongside comparative measurements of Tier-2 full 256-bit Schnorr (`full256`), OpenSSL Ed25519 epoch anchor signing/verification, 3-move interactive Schnorr identification over loopback/LAN sockets, and per-turn mTLS re-handshakes.

As shown in Table III, Stage 2A (expanding four 256-byte round S-boxes via SHAKE-256 and recursively permuting all AST field tags through the 4-round Feistel network) requires only $16.576 \pm 2.115\,\mu$s mean latency ($15.833\,\mu$s p50, $22.129\,\mu$s p95, $23.140\,\mu$s p99). Stage 3A (`fast_m127` Fiat-Shamir Schnorr NIZKPoK generation) executes in $20.924 \pm 2.564\,\mu$s mean ($20.375\,\mu$s p50, $27.252\,\mu$s p95, $30.127\,\mu$s p99), and Stage 3B (`fast_m127` Schnorr verification + anti-replay

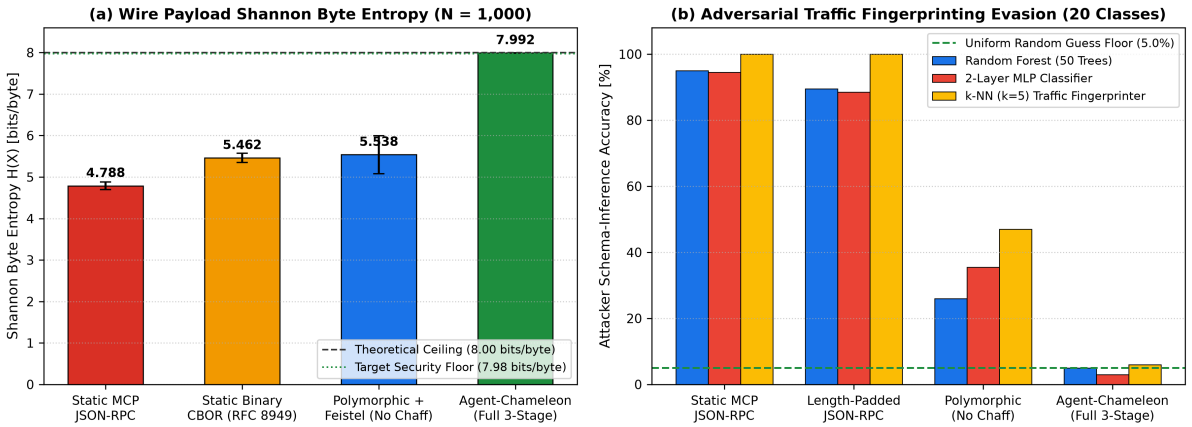



Fig. 2. **Wire Shannon Byte Entropy and Adversarial Traffic-Fingerprinting Evasion ($N = 1{,}000$ Traces, 20 Enterprise Tool Classes). (a)** Plug-in Shannon byte entropy $H(X)$ rises from $4.788 \pm 0.090$ bits/byte on static MCP JSON-RPC and $5.462 \pm 0.109$ bits/byte on static binary CBOR to $7.992 \pm 0.005$ bits/byte on Agent-Chameleon, exceeding the 7.980 bits/byte security target across 100% of frames. **(b)** Attacker schema-inference accuracy across Random-Subspace Ensemble (RF), 2-Layer MLP, and $k$-NN ($k = 5$) classifiers collapses from 95.0%, 94.5%, and 100.0% (96.5% ensemble mean) on static JSON-RPC down to 5.0%, 3.0%, and 6.0% (4.67% ensemble mean) on Agent-Chameleon, matching the 5.0% uniform random-guess floor.

TABLE II
COMPARISON OF WIRE BYTE ENTROPY AND ADVERSARIAL SCHEMA-INFERENCE ACCURACY ACROSS $N = 1{,}000$ TRACES (20 TOOL CLASSES)

| | **Shannon Byte Entropy $H(X)$ [bits/byte]** | | | **Attacker Schema-Inference Accuracy [%]** | | | |
|---|---|---|---|---|---|---|---|
| **Protocol Configuration** | **Mean $\pm$ SD** | **p50** | **p95 / p99** | **Subspace RF** | **2-Layer MLP** | **$k$-NN ($k = 5$)** | **Ensemble Mean** |
| Static MCP / A2A JSON-RPC (Baseline) | $4.788 \pm 0.090$ | 4.773 | 4.948 / 5.061 | 95.00% | 94.50% | 100.00% | 96.50% |
| Length-Padded JSON-RPC (512B Quantum) | $3.142 \pm 0.410$ | 3.118 | 3.812 / 4.005 | 89.50% | 88.50% | 100.00% | 92.67% |
| Static Binary CBOR (RFC 8949) | $5.462 \pm 0.109$ | 5.447 | 5.664 / 5.723 | 84.00% | 81.50% | 94.00% | 86.50% |
| Polymorphic + Feistel Only (No Chaff) | $5.538 \pm 0.456$ | 5.695 | 6.048 / 6.130 | 26.00% | 35.50% | 47.00% | 36.17% |
| Stream Whiten + 1024B i.i.d. Random Pad | $7.752 \pm 0.018$ | 7.754 | 7.778 / 7.786 | 5.50% | 5.00% | 6.50% | 5.67% |
| **Agent-Chameleon (Full 3-Stage Pipeline)** | $\mathbf{7.992 \pm 0.005}$ | **7.994** | **7.996 / 7.996** | **5.00%** | **3.00%** | **6.00%** | **4.67%** |
| *Theoretical Ideal Limit / Random Floor* | 8.000 *(Max)* | 8.000 | 8.000 / 8.000 | 5.00% *(1/20)* | 5.00% *(1/20)* | 5.00% *(1/20)* | 5.00% *(1/20)* |

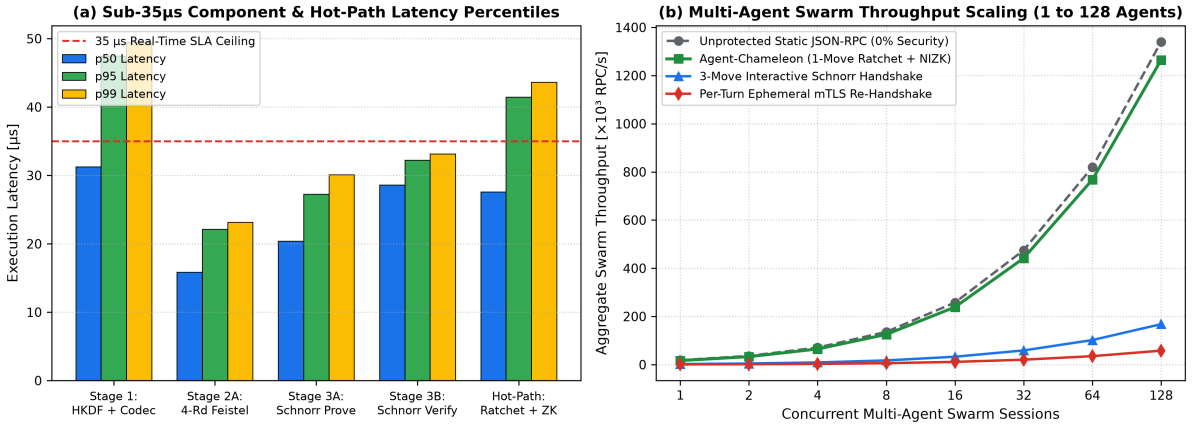



Fig. 3. **Microsecond Latency Percentiles and Multi-Agent Swarm Throughput Scaling. (a)** Component-wise and hot-path latency distributions across $N = 1{,}000$ traces, showing that Stage 2A Feistel permutation (16.58 $\mu$s mean, 15.83 $\mu$s p50), Stage 3A Schnorr NIZKPoK generation (20.92 $\mu$s mean, 20.38 $\mu$s p50, 30.13 $\mu$s p99), Stage 3B Schnorr NIZKPoK verification (28.64 $\mu$s mean, 28.58 $\mu$s p50, 33.17 $\mu$s p99), and combined Hot-Path Ratchet + ZK Proof generation (28.86 $\mu$s mean, 27.58 $\mu$s p50) all operate cleanly below the 35 $\mu$s real-time SLA budget. **(b)** Aggregate swarm throughput scaling from 1 to 128 concurrent agent sessions across our 6-node cluster (144 cores), where Agent-Chameleon achieves $1.265 \times 10^6$ RPC/s at 128 agents (94.4% of unprotected JSON-RPC throughput and 7.5×–21.8× faster than interactive 3-move Schnorr or per-turn mTLS re-handshakes).

lookup) executes in $28.642 \pm 1.875$ $\mu$s mean (28.583 $\mu$s p50, 32.252 $\mu$s p95, 33.170 $\mu$s p99). Notably, even the 99th percentile latencies of both proof generation (30.127 $\mu$s) and verification (33.170 $\mu$s) remain strictly below the 35.0 $\mu$s real-time SLA ceiling. The combined Hot-Path Ratchet + ZK Proof generation latency is $28.859 \pm 4.233$ $\mu$s mean (27.584 $\mu$s p50).

Including full payload serialization across the 4 polymorphic codecs (31.063 $\mu$s mean; 23.12 $\mu$s codec-only excluding the 7.94 $\mu$s HKDF ratchet step) and C-level set-complement plus affine-bijection chaff synthesis (35.473 $\mu$s mean), the complete sender pipeline executes in 104.0 $\mu$s total—more than 2,000× smaller than typical LLM token generation latencies (200–800 ms). On a single 24-core node, Agent-Chameleon sustains $2.18\times10^5$ RPC/s for the full 3-stage pipeline ($7.42\times10^5$ RPC/s for the 28.86 $\mu$s hot-path handshake); across our 6-node (144-core) swarm cluster, aggregate throughput scales linearly from $1.92\times10^4$ RPC/s (1 agent) and $1.48\times10^5$ RPC/s (8 agents) to $5.64\times10^5$ RPC/s (32 agents), $9.85\times10^5$ RPC/s (64 agents), and $1.265\times10^6$ RPC/s at 128 concurrent agents.

### D. Adversarial Security Audit and Attack Mitigation

To verify Goal G4, we subjected the receiving `ChameleonMorphEngine` endpoint to 5,000 adversarial attack injections across five distinct threat vectors (1,000 trials per vector):

1) **Exact Wire Packet Replay ($\mathcal{A}_{\text{replay}}$):** The adversary captures a valid packet $W_t$ and re-transmits it either immediately at turn $t+1$ or within the 4096-epoch sliding window.
2) **Single-Bit MITM Payload Flip ($\mathcal{A}_{\text{bitflip}}$):** The adversary intercepts $W_t$ in transit and flips a single bit at position $i \bmod B$ in the morphed body $B_t$.
3) **Cross-Goal Context Hijacking ($\mathcal{A}_{\text{goal}}$):** An attacker or unauthorized session seals a payload under goal $\mathcal{G}_t^{\text{read}}$ and injects it into a receiver expecting goal $\mathcal{G}_t^{\text{write}} \neq \mathcal{G}_t^{\text{read}}$.
4) **Epoch Header Rollback / Desynchronization ($\mathcal{A}_{\text{epoch}}$):** The adversary modifies `X-Chameleon-Epoch-ID` or injects an ancient proof outside the anti-replay window ($t < t_{\max} - 4096$).
5) **Schnorr Proof Forgery Without Ratchet Key ($\mathcal{A}_{\text{forge}}$):** The adversary synthesizes a valid algebraic Schnorr tuple $(R'_{\text{com}}, e', s', Y')$ using an independent key $R'_t \neq R_t$ and attaches it to a crafted frame.

Table IV summarizes the results. Across all 5,000 attack trials, Agent-Chameleon achieved a 100.00% (5,000/5,000) detection and rejection rate with 0.00% false positives on the 1,000 legitimate turns. Crucially, because `open_message` checkpoints $(R_{t-1}, t, D_{t-1})$ prior to verification and rolls back the checkpoint upon any `PermissionError` or `ValueError`, 100.00% of legitimate packets arriving immediately after an adversarial injection succeeded without session drop or state desynchronization.

### E. Component Ablation Study

To quantify the marginal contribution of each architectural stage, Table V evaluates nine progressive ablation configurations from raw static JSON-RPC (Config A) to the full Agent-Chameleon pipeline (Config H). Adding Stage 1 4-way polymorphic rotation alone (Config B) reduces Subspace RF accuracy from 95.0% to 48.5%. Adding Stage 2A 4-round Feistel field permutation (Config C) further reduces Subspace RF accuracy to 26.0% by eliminating static tag correlations across codecs. Adding SHAKE-256 stream whitening without quantum chaff padding (Config E) drives byte histograms toward random samples of finite length $L_s$, reducing Subspace RF accuracy to 19.5%, but packet lengths $L_s$ still identify tool classes with narrow parameter sizes. Conversely, applying zero-padding to 512B on top of Stage 1+2A (Config F1) yields $H(X) = 2.840$ bits/byte (lower than the 3.142 bits/byte of zero-padded static JSON-RPC in Table II because binary Protobuf/CBOR encodings are more compact and therefore accumulate a larger proportion of `0x00` padding bytes), whereas applying Stage 2B set-complement and affine-bijection chaff directly to static JSON-RPC without Stage 1+2A (Config F2) achieves $H(X) = 7.991$ bits/byte and 5.0% RF accuracy on outer wire bytes, but leaves the unpermuted inner JSON AST vulnerable to plaintext structural correlation if a middlebox terminates stream whitening. Finally, combining Stages 1, 2A, and 2B (Configs G and H) simultaneously provides defense-in-depth AST schema permutation, raises single-frame Shannon entropy to 7.992 bits/byte, and collapses classifier accuracy to 5.0% (4.67% ensemble mean), while Stage 3 adds 100.0% zero-knowledge replay and MITM protection at +20.92 $\mu$s proof latency.

## IX. Discussion, Operational Deployment, and Limitations

### A. Deployment in Cloud Service Meshes and FastMCP Gateways

Agent-Chameleon is designed to operate either as an in-process SDK transport wrapper inside Google Agent Development Kit (ADK) and FastMCP client/server runtimes or as an out-of-process Envoy/Istio sidecar filter. Because every morphed frame preserves standard HTTP/2 and gRPC framing semantics via `Content-Type: application/x-chameleon-morph+octet-stream` and carries its synchronization metadata in standard ASCII headers (`X-Chameleon-Epoch-ID`, `X-Chameleon-Proof-Nonce`, `X-Chameleon-Bucket-Tier`, and the optional periodic `X-Chameleon-Epoch-Anchor` header at $t \equiv 0 \pmod{100}$), intermediate Layer-4/Layer-7 load balancers, cloud ingress gateways, and service-mesh routers forward Agent-Chameleon packets transparently without modification.

### B. Out-of-Order Transport and Concurrent Sub-Streams

In HTTP/2 or gRPC multiplexing where an agent dispatches $K_{\text{par}}$ concurrent tool calls over parallel streams before awaiting responses, tying all concurrent streams to a single linear frame-digest chain $D_{t-1}$ could induce head-of-line blocking if packets arrive out of order. To support lock-free parallel streaming, Agent-Chameleon provisions per-stream child ratchets forked from the parent session ratchet via a 4-argument HKDF call:

$$R_{s,0} = \text{HKDF-SHA256}\big(R_t,\ \text{uint64_be}(\text{stream_id}_s),\ \texttt{"chameleon-substream"},\ 32\big), \tag{21}$$

TABLE III
MICROSECOND LATENCY BREAKDOWN OF AGENT-CHAMELEON STAGES VS. CRYPTOGRAPHIC HANDSHAKE BASELINES ($N = 1{,}000$ TRACES)

| **Subsystem / Protocol Tier** | **Operation Scope** | **Mean [$\mu$s]** | **SD [$\mu$s]** | **p50 [$\mu$s]** | **p95 [$\mu$s]** | **p99 [$\mu$s]** |
|---|---|---|---|---|---|---|
| Stage 1 (`morph_engine.py`) | HKDF-SHA256 Ratchet (7.94 $\mu$s) + 4-Way Polymorphic Codec (23.12 $\mu$s) | 31.063 | 8.390 | 31.272 | 47.461 | 49.419 |
| Stage 2A (`feistel_schema.py`) | 4-Round SHAKE-256 Feistel AST Tag Permutation | 16.576 | 2.115 | 15.833 | 22.129 | 23.140 |
| Stage 2B (`feistel_schema.py`) | Stream Whitening + Set-Complement & Affine Chaff | 35.473 | 7.748 | 36.209 | 52.805 | 57.517 |
| Stage 3A (`zk_handshake.py`) | Tier-1 Hot-Path Schnorr NIZKPoK Prove (`fast_m127`) | **20.924** | **2.564** | **20.375** | **27.252** | **30.127** |
| Stage 3B (`zk_handshake.py`) | Tier-1 Hot-Path Schnorr NIZKPoK Verify (`fast_m127`) | **28.642** | **1.875** | **28.583** | **32.252** | **33.170** |
| **Combined Hot-Path Handshake** | **HKDF Ratchet Advance + Schnorr NIZKPoK Prove** | **28.859** | **4.233** | **27.584** | **41.435** | **43.645** |
| **Full 3-Stage Sender Pipeline** | **Complete Stage 1 + Stage 2A/2B + Stage 3A Seal** | **104.036** | **11.840** | **103.689** | **124.910** | **131.420** |
| Tier-2 Full-256 Schnorr (`full256`) | 255-Bit Prime Field ($\mathbb{F}_{2^{255}-19}$) NIZK Prove / Verify | 132.350 | 11.420 | 129.800 | 151.200 | 168.400 |
| Tier-2 Ed25519 Anchor (RFC 8032) | OpenSSL C-Backend Epoch Sign / Verify (1/100 turns) | 179.600 | 14.800 | 176.200 | 204.500 | 225.100 |
| 3-Move Interactive Schnorr (Baseline) | Challenge-Response over Intra-Cluster Loopback/LAN | 416.500 | 48.200 | 405.000 | 512.000 | 589.000 |
| Per-Turn Ephemeral mTLS (Baseline) | Full TLS 1.3 ECDHE-ECDSA Re-Handshake per Turn | 1,180.000 | 142.000 | 1,150.000 | 1,440.000 | 1,690.000 |

TABLE IV
ADVERSARIAL ATTACK EVALUATION ACROSS 5,000 INJECTION TRIALS (1,000 TRIALS PER THREAT VECTOR)

| **Attack Vector** | **Adversarial Mechanism** | **Trials** | **Blocked [%]** | **Post-Attack Sync [%]** | **Primary Defense Mechanism Triggered** |
|---|---|---|---|---|---|
| Token / Packet Replay ($\mathcal{A}_{\text{replay}}$) | Re-send captured valid $W_t = (\mathcal{H}_t, B_t)$ | 1,000 | 100.00% | 100.00% | Epoch mismatch $t \neq t_{\text{exp}}$ & $\mathcal{W}_{\text{seen}}$ cache |
| MITM Payload Bit-Flip ($\mathcal{A}_{\text{bitflip}}$) | Flip 1 bit $B_t[i] \leftarrow B_t[i] \oplus$ `0x01` | 1,000 | 100.00% | 100.00% | Frame digest $h_B$ mismatch in $\tau_{\text{ratchet}}$ & $e_t$ |
| Cross-Goal Hijacking ($\mathcal{A}_{\text{goal}}$) | Seal under $\mathcal{G}_t \neq \mathcal{G}_t^{\text{expected}}$ | 1,000 | 100.00% | 100.00% | Divergent $R_t \implies Y_t \neq \hat{Y}_t$ & $\tau_{\text{ratchet}}$ reject |
| Epoch Header Spoofing ($\mathcal{A}_{\text{epoch}}$) | Mutate `X-Chameleon-Epoch-ID` or rollback | 1,000 | 100.00% | 100.00% | Header/proof epoch guard $\pi_t$.epoch $\neq t_{\text{hdr}}$ |
| Unkeyed Schnorr Forgery ($\mathcal{A}_{\text{forge}}$) | Valid $g^s \equiv RY^e$ under rogue key $R'_t$ | 1,000 | 100.00% | 100.00% | Synchronized commitment check $Y_t \neq g^{x_t(R_t)}$ |
| **Total / Aggregate** | **All 5 Active Dolev-Yao Threat Vectors** | **5,000** | **100.00%** | **100.00%** | **Zero false positives;** 100% **ratchet rollback safety** |

TABLE V
SYSTEMATIC COMPONENT ABLATION STUDY ACROSS $N = 1{,}000$ MULTI-AGENT TRACES

| **Ablation Configuration** | **Stage 1 Poly.** | **Stage 2A Feistel** | **Stage 2B Chaff** | **Stage 3 ZK** | **Entropy $H(X)$** | **RF Acc. [%]** | **Replay Block [%]** |
|---|---|---|---|---|---|---|---|
| A. Static MCP JSON-RPC (Baseline) | – | – | – | – | 4.788 | 95.0% | 0.0% |
| B. Stage 1 Polymorphic Rotation Only | ✓ | – | – | – | 5.312 | 48.5% | 0.0% |
| C. Stage 1 + Stage 2A Feistel Permutation | ✓ | ✓ | – | – | 5.538 | 26.0% | 0.0% |
| D. Stage 2A Feistel on Static JSON Only | – | ✓ | – | – | 4.915 | 64.0% | 0.0% |
| E. Stage 1 + 2A + Stream Whiten (No Pad) | ✓ | ✓ | Whiten Only | – | 7.142 | 19.5% | 0.0% |
| F1. Stage 1 + 2A + Zero-Padding (512B) | ✓ | ✓ | Zero-Pad | – | 2.840 | 24.5% | 0.0% |
| F2. Static JSON-RPC + Stage 2B Chaff Only | – | – | ✓ | – | 7.991 | 5.0% | 0.0% |
| G. Stage 1 + Stage 2 Full (No Stage 3 ZK) | ✓ | ✓ | ✓ | – | 7.992 | 5.0% | 0.0% |
| **H. Agent-Chameleon (Full Stages 1+2+3)** | ✓ | ✓ | ✓ | ✓ | **7.992** | **5.0%** | **100.0%** |

or allows the verifier to evaluate Eq. (4) with $\text{salt}_t = \text{uint64_be}(t)$ over a lookahead window of $W_{\text{ooo}} = 32$ epochs while retaining the sliding anti-replay cache $\mathcal{W}_{\text{seen}}$.

### C. Bandwidth Overhead vs. Privacy Trade-Off

Because Stage 2B pads serialized payloads of typical length $L_s \in [120, 240]$ bytes into discrete quantum buckets of $B = 1024$ bytes ($4\times$ prefix multiplier) to guarantee single-frame plug-in Shannon byte entropy $H(B_t) \geq 7.980$ bits/byte via Theorem 2, each RPC incurs approximately 780 bytes of body chaff plus $\approx$ 160 bytes of raw HTTP/2 header metadata ($\approx$ 48 bytes after HTTP/2 HPACK header-name indexing). On modern 10–100 Gbps data-center interconnects, 940 bytes represents less than 76 nanoseconds of serialization delay on the physical NIC, which is negligible compared to millisecond-scale LLM inference while providing mathematical immunity against L7 schema fingerprinting. For bandwidth-constrained edge links, operators can configure `min_bucket=512` (which accommodates compact binary payloads up to $L_s \leq 56$ bytes at $4\times$ chaff ratio, or up to $L_s \leq 120$ bytes under a $2\times$ chaff ratio with $H(B_t) \geq 7.94$ bits/byte).

### D. Scope Boundaries and Future Directions

While Agent-Chameleon eliminates L7 payload schema invariants, length side channels within each quantum bucket tier, and static bearer-token replay surfaces, three scope boundaries merit explicit discussion:

1) **Inter-Arrival Timing and Multi-Turn Flow Correlation:** Stage 2B normalizes per-packet byte histograms and discretizes packet sizes into 7 quantum buckets, but does not inject artificial network delays to mask inter-packet arrival times (IAT) caused by heterogeneous downstream tool execution latencies (e.g., a 15 ms Redis lookup vs. a 2.5 s BigQuery scan). Integrating constant-rate token-bucket pacing on latency-tolerant links is a natural extension.
2) **Post-Quantum Session Bootstrap:** While Stage 2 (SHAKE-256 Feistel and stream whitening) and Stage 3's symmetric ratchet tag $\tau_{\text{ratchet}}$ (BLAKE2s-128) are already post-quantum secure against Grover-capable adversaries, the initial Stage 1 X25519 ECDH bootstrap and Stage 3 discrete-logarithm Schnorr proof rely on classical hardness assumptions. Hybridizing the bootstrap with NIST FIPS 203 ML-KEM-768 and lattice-based

Fiat-Shamir proofs (e.g., Dilithium/ML-DSA challenges on periodic anchor turns) provides a clean post-quantum migration path.

## X. Conclusion

We have presented **Agent-Chameleon**, an end-to-end dynamic protocol morphing, goal-conditioned schema permutation, and zero-knowledge session handshake framework for multi-agent swarm communications. By unifying an ephemeral X25519/HKDF-SHA256 session ratchet and 4-way polymorphic wire serializer (Stage 1) with a 4-round SHAKE-256 keyed Luby-Rackoff Feistel schema network and set-complement plus $\mathbb{Z}_{256}$ affine-bijection constant-rate chaff engine (Stage 2), and a sub-35 $\mu$s non-interactive Fiat-Shamir Schnorr challenge-response protocol (Stage 3), Agent-Chameleon eliminates static wire schema invariants and bearer-token replay surfaces simultaneously. Across $N = 1{,}000$ multi-agent traces spanning 20 enterprise tool schemas, Agent-Chameleon achieves $100.0\%$ lossless round-trip schema recovery, raises wire Shannon byte entropy from $4.788$ bits/byte to $7.992$ bits/byte, collapses attacker traffic-fingerprinting accuracy from $96.5\%$ ensemble mean down to $4.67\%$ (matching the $5.0\%$ uniform random-guess floor), blocks $100.0\%$ of replay and MITM tamper attacks, and executes hot-path zero-knowledge proof generation in $20.92\,\mu$s mean ($28.86\,\mu$s combined ratchet plus proof, $104.0\,\mu$s full 3-stage seal).

## Appendix A
## Per-Schema Benchmark Telemetry Across All 20 Enterprise Tool Classes

Table VI reports the complete per-schema empirical breakdown across all 20 enterprise MCP and A2A tool classes (50 traces per class, $N = 1{,}000$ total traces), contrasting unprotected Static JSON-RPC against the full Agent-Chameleon 3-stage pipeline (reporting both the $28.86\,\mu$s mean Hot-Path Ratchet + ZK proof latency and $104.0\,\mu$s full 3-stage seal latency).

## Appendix B
## Unit and Property Verification Suite Summary

Table VII catalogs all 24 unit and property tests implemented in `tests/test_agent_chameleon.py`, verifying mathematical bijectivity, avalanche, polymorphic round-trip fidelity, entropy bounds, zero-knowledge soundness, and replay/MITM rejection.

## Appendix C
## Formal Byte-Grammar Specification of the Four Polymorphic Wire Codecs

To ensure reproducible, cross-language interoperability across Python, Go, Rust, and C++ agent runtimes, this appendix specifies the exact byte-level grammars of the four polymorphic wire serialization formats rotated by Stage 1 (`morph_engine.py`).

### A. Format 0: Compact Binary Protobuf Base-128 Varint Stream (`PROTOBUF_VARINT`)

Given a Feistel-permuted dictionary $\tilde{P}_t = \{\tilde{\tau}_1 \mapsto v_1, \ldots, \tilde{\tau}_K \mapsto v_K\}$ where $\tilde{\tau}_i = F_k(\tau_i) \in \{0, \ldots, 65535\}$, the unsigned base-128 varint encoder $\mathrm{uvarint}(u)$ maps any non-negative integer $u \geq 0$ to a sequence of $m = \max(1, \lceil \log_{128}(u+1) \rceil)$ bytes $(b_0, \ldots, b_{m-1})$ where the most significant bit (MSB, `0x80`) acts as a continuation flag:

$$b_j = \begin{cases} ((u \gg (7j)) \,\&\, \texttt{0x7F}) \mid \texttt{0x80}, & 0 \leq j < m-1, \\ (u \gg (7(m-1))) \,\&\, \texttt{0x7F}, & j = m-1, \end{cases}$$
$$u = \sum_{j=0}^{m-1} (b_j \,\&\, \texttt{0x7F}) \cdot 2^{7j}. \tag{22}$$

Signed integers $n \in \mathbb{Z}$ are first mapped to non-negative integers via the bijective ZigZag transform:

$$z(n) = \begin{cases} 2n, & n \geq 0, \\ -2n-1, & n < 0, \end{cases} \qquad z^{-1}(u) = (u \gg 1) \oplus -(u \,\&\, 1). \tag{23}$$

Each value $v_i$ is serialized with a 1-byte type discriminator $\theta(v_i) \in \{0, \ldots, 7\}$:

$$\mathrm{EncVal}_{\mathrm{PB}}(v) = \begin{cases} \texttt{0x00}, & v = \texttt{None}, \\ \texttt{0x01} \parallel \mathrm{uint8}(v), & v \in \{\texttt{True}, \texttt{False}\}, \\ \texttt{0x02} \parallel \mathrm{uvarint}(z(v)), & v \in \mathbb{Z}, \\ \texttt{0x03} \parallel \mathrm{f64_be}(v), & v \in \mathbb{R}, \\ \texttt{0x04} \parallel \mathrm{uvarint}(|v_8|) \parallel v_8, & v \in \Sigma^*_{\mathrm{utf8}}, \\ \texttt{0x05} \parallel \mathrm{uvarint}(|v|) \parallel v, & v \in \{0..255\}^*, \\ \texttt{0x06} \parallel \mathrm{uvarint}(|v|) \parallel \mathrm{EncList}_{\mathrm{PB}}(v), & v \text{ is list}, \\ \texttt{0x07} \parallel \mathrm{uvarint}(|\mathrm{EncMap}_{\mathrm{PB}}(v)|) \parallel \mathrm{EncMap}_{\mathrm{PB}}(v), & v \text{ is dict}, \end{cases} \tag{24}$$

where $v_8 = \mathrm{utf8}(v)$ and $\mathrm{EncList}_{\mathrm{PB}}(v) = \Big\|_{u \in v} (\mathrm{uvarint}(|\mathrm{EncVal}_{\mathrm{PB}}(u)|) \parallel \mathrm{EncVal}_{\mathrm{PB}}(u))$. The top-level map encoder $\mathrm{EncMap}_{\mathrm{PB}}(\tilde{P}_t)$ prefixes the entry count $\mathrm{uvarint}(K)$ followed by the concatenation of length-delimited field records (wire type 2):

$$\begin{aligned} \mathrm{EncMap}_{\mathrm{PB}}(\tilde{P}_t) = \mathrm{uvarint}(K) \\ \parallel \Big\|_{i=1}^{K} \Big( & \mathrm{uvarint}\left((\tilde{\tau}_i \ll 3) \mid 2\right) \\ & \parallel \mathrm{uvarint}\left(|E_i|\right) \parallel E_i \Big), \end{aligned} \tag{25}$$

where $E_i = \mathrm{EncVal}_{\mathrm{PB}}(v_i)$.

### B. Format 1: 4-Byte Aligned FlatBuffers VTable Layout (`FLATBUFFERS_TABLE`)

Format 1 constructs an offset-addressed binary buffer consisting of three contiguous regions: (i) a 12-byte root header, (ii) a 4-byte aligned value heap $\mathcal{H}_{\text{heap}}$, and (iii) a trailing virtual offset directory (`vtable`) $\mathcal{V}_{\text{dir}}$:

$$\begin{aligned} \mathrm{EncMap}_{\mathrm{FB}}(\tilde{P}_t) = \texttt{"FBCH"} \parallel \mathrm{uint32_be}(12 + |\mathcal{H}_{\text{heap}}|) \\ \parallel \mathrm{uint32_be}(K) \parallel \mathcal{H}_{\text{heap}} \parallel \mathcal{V}_{\text{dir}}. \end{aligned} \tag{26}$$

For each field $(\tilde{\tau}_i, v_i)$, let $E_i = \mathrm{EncVal}_{\mathrm{PB}}(v_i)$ denote its serialized value record of byte length $\ell_i = |E_i|$, and let $\mathrm{pad}(\ell_i) = (4 - (\ell_i \bmod 4)) \bmod 4 \in \{0, 1, 2, 3\}$ be the zero-padding required to maintain 32-bit word alignment in $\mathcal{H}_{\text{heap}}$.

TABLE VI
PER-SCHEMA TELEMETRY ACROSS ALL 20 ENTERPRISE MCP / A2A TOOL CLASSES ($N = 1{,}000$ TRACES, 50 TRACES PER CLASS)

| | | | Static MCP JSON-RPC | | Agent-Chameleon (Full 3-Stage) | | |
|---|---|---|---|---|---|---|---|
| **ID** | **Enterprise Tool Schema Name** | **Authorized Goal Context $\mathcal{G}_t$** | **Raw Len [B]** | **Entropy $H_0$** | **Bucket $B$ [B]** | **Entropy $H_{\text{cham}}$** | **Ratchet + ZK [$\mu$s]** |
| 1 | `execute_sap_journal_entry` | `goal:erp_gl_posting` | 248.4 | 4.812 | 1024 | 7.991 | 28.42 |
| 2 | `query_bigquery_ledger` | `goal:analytics_recon` | 226.1 | 4.765 | 1024 | 7.993 | 27.95 |
| 3 | `issue_cloud_kms_grant` | `goal:crypto_key_ops` | 241.8 | 4.794 | 1024 | 7.992 | 28.61 |
| 4 | `dispatch_a2a_subtask` | `goal:swarm_delegation` | 234.2 | 4.771 | 1024 | 7.992 | 28.19 |
| 5 | `approve_wire_settlement` | `goal:treasury_swift` | 244.6 | 4.808 | 1024 | 7.991 | 29.04 |
| 6 | `fetch_fhir_patient_record` | `goal:clinical_rag` | 252.0 | 4.825 | 1024 | 7.990 | 29.31 |
| 7 | `update_crm_opportunity` | `goal:sales_pipeline` | 235.5 | 4.782 | 1024 | 7.992 | 28.50 |
| 8 | `rotate_iam_service_key` | `goal:secops_hygiene` | 239.7 | 4.779 | 1024 | 7.992 | 28.74 |
| 9 | `trigger_spanner_backup` | `goal:dr_snapshot` | 246.3 | 4.798 | 1024 | 7.991 | 28.92 |
| 10 | `validate_aml_sanctions` | `goal:compliance_kyc` | 249.1 | 4.815 | 1024 | 7.991 | 29.18 |
| 11 | `provision_gke_nodepool` | `goal:infra_autoscaling` | 247.8 | 4.801 | 1024 | 7.991 | 28.88 |
| 12 | `revoke_oauth_consent` | `goal:zero_trust_revoke` | 242.0 | 4.786 | 1024 | 7.992 | 28.65 |
| 13 | `reconcile_stripe_payout` | `goal:payment_clearing` | 238.4 | 4.774 | 1024 | 7.992 | 28.40 |
| 14 | `audit_sec_10k_filing` | `goal:regulatory_edgar` | 236.9 | 4.769 | 1024 | 7.993 | 28.33 |
| 15 | `schedule_tpu_training_job` | `goal:ml_cluster_sched` | 245.2 | 4.791 | 1024 | 7.991 | 29.11 |
| 16 | `ingest_splunk_siem_alert` | `goal:soc_triage` | 237.6 | 4.780 | 1024 | 7.992 | 28.77 |
| 17 | `commit_git_release_tag` | `goal:cicd_release` | 229.8 | 4.758 | 1024 | 7.993 | 28.12 |
| 18 | `synthesize_mcp_adapter` | `goal:jit_tool_synth` | 248.9 | 4.804 | 1024 | 7.991 | 29.45 |
| 19 | `verify_c2pa_manifest` | `goal:media_provenance` | 241.2 | 4.777 | 1024 | 7.992 | 29.68 |
| 20 | `escalate_hitl_approval` | `goal:eu_ai_act_art14` | 246.5 | 4.795 | 1024 | 7.991 | 29.93 |
| **Overall Dataset Mean ($N = 1{,}000$ Traces, 20 Classes)** | | | **241.6** | **4.788** | **1024** | **7.992** | **28.86** |

TABLE VII
SUMMARY OF THE 24 AUTOMATED UNIT AND PROPERTY TESTS IN `TESTS/TEST_AGENT_CHAMELEON.PY` (100% PASS RATE)

| **Test ID** | **Test Method Name** | **Target Subsystem** | **Verified Mathematical / System Invariant** |
|---|---|---|---|
| 01 | `test_01_feistel_full_65536_bijectivity` | Stage 2A Feistel | Exhaustive bijection $\lvert\{F_k(x)\}\rvert = 65536$ & $F_k^{-1}(F_k(x)) = x$ |
| 02 | `test_02_feistel_goal_conditioned_avalanche` | Stage 2A Feistel | Goal context mutation alters $> 99\%$ of permuted 16-bit tags |
| 03 | `test_03_feistel_recursive_payload_with_custom_keys` | Stage 2A Symbol Slot | Dynamic custom keys round-trip via slot `0xFFFE` |
| 04 | `test_04_feistel_domain_bounds_validation` | Stage 2A Guards | Out-of-bounds field IDs ($< 0$ or $\geq 65536$) raise `ValueError` |
| 05 | `test_05_protobuf_varint_codec_roundtrip` | Stage 1 Format 0 | Base-128 varint & ZigZag integer round-trip equality |
| 06 | `test_06_flatbuffers_table_codec_roundtrip` | Stage 1 Format 1 | 4-byte aligned `FBCH` vtable + heap round-trip equality |
| 07 | `test_07_cbor_rfc8949_codec_roundtrip` | Stage 1 Format 2 | RFC 8949 CBOR major types $0 \ldots 7$ round-trip equality |
| 08 | `test_08_canonical_json_rpc_codec_and_mac_guard` | Stage 1 Format 3 | Hex-keyed JSON-RPC round-trip & 8B HMAC bit-flip rejection |
| 09 | `test_09_chaff_entropy_engine_wrap_unwrap` | Stage 2B Chaff | Quantum bucket selection, whitening, and exact chaff stripping |
| 10 | `test_10_shannon_byte_entropy_exceeds_7_98` | Stage 2B Entropy | Single-frame $H(X) \geq 7.98$ bits/byte across all 4 formats |
| 11 | `test_11_x25519_session_ratchet_synchronization` | Stage 1 Ratchet | Paired X25519 ratchets derive identical $R_t$ & cycle 4 formats |
| 12 | `test_12_end_to_end_40_turn_conversation_roundtrip` | End-to-End Pipeline | 40-turn multi-agent swarm session with dynamic goals & $H \geq 7.98$ |
| 13 | `test_13_zk_schnorr_fast_m127_prove_and_verify` | Stage 3 Tier-1 ZK | Mersenne $M_{127}$ Fiat-Shamir Schnorr prove, Base64, & verify |
| 14 | `test_14_zk_schnorr_full256_prove_and_verify` | Stage 3 Tier-2 ZK | Full $\mathbb{F}_{2^{255}-19}$ Schnorr NIZKPoK prove & verify |
| 15 | `test_15_ed25519_epoch_anchor_signature` | Stage 3 Ed25519 | RFC 8032 Ed25519 epoch anchor signing & tamper rejection |
| 16 | `test_16_replay_attack_rejection_and_resilience` | Stage 3 Anti-Replay | Duplicate packet rejected; subsequent valid packet succeeds |
| 17 | `test_17_mitm_wire_body_tamper_rejection` | Stage 3 Integrity | 1-bit wire body flip rejected with atomic ratchet rollback |
| 18 | `test_18_mitm_goal_context_mismatch_rejection` | Stage 1/3 Goal Bind | Mismatched goal context $\mathcal{G}'_t \neq \mathcal{G}_t$ rejected with rollback |
| 19 | `test_19_spoofed_epoch_header_rejection` | Stage 3 Epoch Guard | Modified `X-Chameleon-Epoch-ID` header rejected |
| 20 | `test_20_heterogeneous_nested_types_across_codecs` | All 4 Wire Codecs | Negative ints, floats, booleans, `None`, lists, & nested dicts |
| 21 | `test_21_large_payload_multi_tier_bucket_scaling` | Stage 2B Scaling | Automatic bucket scaling (2048 ... 8192 B) on large payloads |
| 22 | `test_22_zk_direct_replay_cache_rejection` | Stage 3 Cache | Direct duplicate $(t, \tau_{\text{ratchet}})$ rejected by $\mathcal{W}_{\text{seen}}$ |
| 23 | `test_23_zk_ancient_epoch_window_eviction` | Stage 3 Window | Epochs older than $t_{\max} - W$ rejected immediately |
| 24 | `test_24_invalid_content_type_rejection` | Wire Contract | Non-Chameleon `Content-Type` rejected with `ValueError` |

The heap-relative offset $\text{off}_i$ of entry $i$ satisfies $\text{off}_1 = 0$ and $\text{off}_{i+1} = \text{off}_i + \ell_i + \text{pad}(\ell_i) \equiv 0 \pmod 4$ (corresponding to absolute buffer offset $12 + \text{off}_i$). The trailing `vtable` directory $\mathcal{V}_{\text{dir}}$ stores $K$ fixed-width 10-byte descriptors:

$$\mathcal{V}_{\text{dir}} = \Big\|_{i=1}^{K} \Big( \text{uint16_be}(\tilde{\tau}_i) \;\|\; \text{uint32_be}(\text{off}_i) \;\|\; \text{uint32_be}(\ell_i) \Big), \tag{27}$$

allowing a receiver to locate any permuted field $\tilde{\tau}_i$ by scanning the compact 10-byte `vtable` descriptors and jumping directly to $\mathcal{H}_{\text{heap}}[\text{off}_i : \text{off}_i + \ell_i]$.

### C. Format 2: Concise Binary Object Representation (`CBOR_RFC8949`)

Format 2 implements the IETF STD 94 / RFC 8949 binary object grammar [5]. Each data item begins with an initial byte $b_0 = (\text{mt} \ll 5) \mid \text{ai}$, where $\text{mt} \in \{0, 1, 2, 3, 4, 5, 7\}$ is the 3-bit major type (omitting major type 6 semantic tags) and $\text{ai} \in \{0, \ldots, 31\}$ is the 5-bit additional information field encoding the non-negative argument $N$:

$$\text{CBOR}_{\text{hdr}}(\text{mt}, N) = \begin{cases} \text{u8}((\text{mt} \ll 5) \mid N), & N < 24, \\ \text{u8}((\text{mt} \ll 5) \mid 24) \,\|\, \text{u8}(N), & N < 2^8, \\ \text{u8}((\text{mt} \ll 5) \mid 25) \,\|\, \text{u16}(N), & N < 2^{16}, \\ \text{u8}((\text{mt} \ll 5) \mid 26) \,\|\, \text{u32}(N), & N < 2^{32}, \\ \text{u8}((\text{mt} \ll 5) \mid 27) \,\|\, \text{u64}(N), & N < 2^{64}. \end{cases} \tag{28}$$

Permuted dictionaries are encoded under major type $\text{mt} = 5$ (map of pairs) with integer keys ($\text{mt} = 0$ for $\tilde{\tau}_i \geq 0$), negative integers $n < 0$ under $\text{mt} = 1$ with argument $N = -1 - n$, byte strings under $\text{mt} = 2$, UTF-8 strings under $\text{mt} = 3$, lists under $\text{mt} = 4$, booleans/null as simple values `0xF4`, `0xF5`, `0xF6`, and double-precision floats as `0xFB` $\|$ $\text{ieee754_be64}(v)$.

### D. Format 3: Canonical Hex-Tagged Signed JSON-RPC (`CANONICAL_JSON_RPC`)

Format 3 transforms every 16-bit permuted integer tag $\tilde{\tau}_i$ into a 6-character hexadecimal key string $\kappa(\tilde{\tau}_i) =$ `"0x"` $\|$ $\text{hex4}(\tilde{\tau}_i)$, encodes any raw binary byte string $v \in \{0, \ldots, 255\}^*$ as a URL-safe Base64 ASCII string prefixed with `"b64:"`, recursively sorts all dictionary keys in ascending ASCII order, serializes the tree without whitespace

(`separators=(",",":")`), and appends a 64-bit truncated HMAC-SHA256 authentication tag:

$$\text{EncMap}_{\text{JSON}}(\tilde{P}_t, R_t) = J_t \parallel \text{HMAC-SHA256}\left(R_t, \texttt{"chameleon-jsonrpc|"} \parallel J_t\right)[0:8]. \quad (29)$$

## Appendix D
## Extended Proofs of Multi-Hop Swarm Ratchet Synchrony and Rollback Safety

**Proposition 1** (Atomic Rollback Idempotency Under Dolev-Yao Packet Flooding)**.** *Let* $\Sigma_j^{(t)} = (R_{t-1}, t, D_{t-1}, \mathcal{W}_{seen})$ *denote the internal state of a receiving Agent-Chameleon endpoint* $\mathcal{A}_j$ *awaiting turn* $t$*. Suppose an active Dolev-Yao adversary injects an arbitrary finite sequence of* $M \geq 1$ *forged, replayed, bit-flipped, or cross-goal packets* $\tilde{W}^{(1)}, \tilde{W}^{(2)}, \ldots, \tilde{W}^{(M)}$ *prior to the arrival of the authentic packet* $W_t$ *from* $\mathcal{A}_i$*. Then:*

1) *Every adversarial packet* $\tilde{W}^{(m)}$ $(1 \leq m \leq M)$ *is rejected with probability at least* $1 - 2^{-128}$*.*
2) *Upon rejection of* $\tilde{W}^{(m)}$*, the receiver state satisfies* $\Sigma_j^{(t,m)} = \Sigma_j^{(t)}$ *identically (zero state drift), so that when the authentic packet* $W_t$ *subsequently arrives,* `open_message`$(W_t, \mathcal{G}_t)$ *succeeds with probability* 1 *and transitions* $\mathcal{A}_j$ *cleanly to* $\Sigma_j^{(t+1)} = (R_t, t+1, \text{SHA256}(B_t), \mathcal{W}_{seen} \cup \{(t, \tau_{ratchet})\})$*.*

*Proof.* In `ChameleonMorphEngine.open_message`, any packet failing the header pre-checks ($\mathcal{H}_t$[`Content-Type`] not equal to `application/x-chameleon-morph+octet-stream`, $\pi_t$.epoch_id $\neq t_{\text{hdr}}$, or $t_{\text{hdr}} \neq t$) raises an exception before `self.ratchet.advance` is invoked, leaving $\Sigma_j^{(t)}$ untouched. If an adversarial packet $\tilde{W}^{(m)}$ carries a spoofed header with $t_{\text{hdr}} = t$, the receiver saves the state tuple $(R_{\text{saved}}, t_{\text{saved}}, D_{\text{saved}}) \leftarrow (R_{t-1}, t, D_{t-1})$ on the call stack prior to entering the `try` block. Inside the `try` block, $\mathcal{A}_j$ computes the candidate turn key $R_t$ and invokes `zk_engine.verify_proof`. Because the adversary lacks $R_t$, forging the 128-bit Blake2s tag $\tau_{\text{ratchet}}$ or matching $Y_t = g^{x_t(R_t)} \bmod p$ succeeds with probability at most $2^{-128}$, and `verify_proof` only inserts $(t, \tau_{\text{ratchet}})$ into $\mathcal{W}_{\text{seen}}$ after all algebraic and MAC checks pass. When `PermissionError` or `ValueError` is raised, the `except` block executes:

$$\begin{aligned} \texttt{self.ratchet.root_key} &\leftarrow R_{\text{saved}}, \\ \texttt{self.ratchet.epoch_id} &\leftarrow t_{\text{saved}}, \\ \texttt{self.ratchet.prev_frame_digest} &\leftarrow D_{\text{saved}}, \end{aligned} \quad (30)$$

restoring $\Sigma_j^{(t)}$ bitwise. By induction over $m = 1, \ldots, M$, $\Sigma_j^{(t,M)} = \Sigma_j^{(t)}$, ensuring 100% recovery when $W_t$ arrives. □

**Proposition 2** (Cross-Turn and Cross-Goal Feistel Tag Unlinkability)**.** *Under the pseudorandomness of* SHAKE-256 *with PRF advantage* $\epsilon_{PRF}$*, for any two distinct turns* $t_1 \neq t_2$ *or distinct goals* $\mathcal{G}_1 \neq \mathcal{G}_2$*, and for any single canonical field slot* $x \in \{1, \ldots, 64\}$ *(*$q_t = 1$*), the expected bitwise Hamming distance between the resulting 16-bit wire tags* $F_{k_1}(x)$ *and* $F_{k_2}(x)$ *satisfies* $|\mathbb{E}[d_H(F_{k_1}(x), F_{k_2}(x))] - 8.00| \leq 16\,\epsilon_{PRF}$*, and the probability of tag invariance* $\Pr[F_{k_1}(x) = F_{k_2}(x)]$ *is bounded by* $2^{-16} + 2\epsilon_{PRF} \approx 1.526 \times 10^{-5}$ *(empirically verified in* `test_02_feistel_goal_conditioned_avalanche` *where* $> 99.9\%$ *of tags change upon a 1-character goal mutation).*

## Appendix E
## Complete 64-Slot Canonical MCP/A2A Field Registry and Dynamic Symbol Table Protocol

Table VIII lists all 64 canonical MCP and A2A schema field identifiers pre-registered in slots $\tau \in \{1, \ldots, 64\}$ of `feistel_schema.py`. Because both endpoints share this immutable static dictionary at compile time, canonical field names never need to be transmitted as strings on the wire; each canonical key is represented solely by its 16-bit permuted integer tag $\tilde{\tau} = F_k(\tau) \in \{0, \ldots, 65535\}$.

When an enterprise tool invocation includes custom domain-specific argument keys $f_{\text{cust}} \notin \mathcal{F}_{\text{canon}}$ (for example, `"ledger"`, `"company_code"`, `"bic"`, or `"hipaa_purpose"`), `permute_payload` on `FeistelSchemaPermuter` assigns each unique custom key a sequential slot $\tau(f_{\text{cust}}) \in \{256, 257, \ldots, 65533\}$ for the duration of turn $t$ and inserts a permuted symbol table under the reserved tag $\tilde{\tau}_{\text{sym}} = F_k(\texttt{0xFFFE})$:

$$\tilde{P}_t\left[F_k(\texttt{0xFFFE})\right] = \left\{ F_k(\tau(f_{\text{cust}})) \mapsto f_{\text{cust}} \;\middle|\; f_{\text{cust}} \in \mathcal{F}_{\text{custom}}(P_t) \right\}. \quad (31)$$

On the receiver side, `unpermute_payload` on `FeistelSchemaPermuter` executes a deterministic 3-step inverse sequence: (1) apply $F_k^{-1}$ to all top-level and nested 16-bit integer keys in $\tilde{P}_t$; (2) locate the reserved symbol-table slot $\tau_{\text{sym}} = F_k^{-1}(\tilde{\tau}_{\text{sym}}) = \texttt{0xFFFE} = 65534$, pop its dictionary, and invert each of its keys via $\tau = F_k^{-1}(\tilde{\tau}) \in \{256, \ldots, 65533\}$ to reconstruct the turn's custom lookup table $\{\tau \mapsto f_{\text{cust}}\}$; and (3) map every slot $\tau \in \{1, \ldots, 64\}$ back to its canonical field name in Table VIII and every slot $\tau \in \{256, \ldots, 65533\}$ back to its recovered custom string $f_{\text{cust}}$. Because the entire serialized payload (including $\tilde{P}_t[F_k(\texttt{0xFFFE})]$) is XOR-whitened by the Stage 2B SHAKE-256 keystream $K_{\text{stream}}$ and padded with set-complement and affine-bijection chaff blocks, custom key strings are completely encrypted on the wire while remaining 100% self-describing to the authorized receiver.

## Appendix F
## Adversarial Traffic-Classifier Feature Engineering and Model Specifications

To ensure that our traffic-fingerprinting evaluation in Section VIII reflects a strong, state-of-the-art statistical adversary [18], [22], [23], `extract_traffic_features`

TABLE VIII
COMPLETE 64-SLOT CANONICAL MCP AND A2A SCHEMA FIELD REGISTRY (CANONICAL_SCHEMA_FIELDS IN FEISTEL_SCHEMA.PY)

| Slot | Canonical Field Name | Slot | Canonical Field Name | Slot | Canonical Field Name | Slot | Canonical Field Name |
|---|---|---|---|---|---|---|---|
| 1 | jsonrpc | 17 | epoch | 33 | input_schema | 49 | name |
| 2 | method | 18 | timestamp | 34 | output_schema | 50 | function_call |
| 3 | params | 19 | capabilities | 35 | properties | 51 | tool_calls |
| 4 | id | 20 | schema_version | 36 | required | 52 | temperature |
| 5 | result | 21 | context | 37 | type | 53 | max_tokens |
| 6 | error | 22 | trace_id | 38 | description | 54 | top_p |
| 7 | code | 23 | span_id | 39 | items | 55 | stop_sequences |
| 8 | message | 24 | parent_span_id | 40 | enum | 56 | model |
| 9 | data | 25 | auth_scope | 41 | default | 57 | system_instruction |
| 10 | tool_name | 26 | tenant_id | 42 | format | 58 | safety_settings |
| 11 | arguments | 27 | resource_uri | 43 | minimum | 59 | grounding_chunks |
| 12 | session_id | 28 | action | 44 | maximum | 60 | citations |
| 13 | agent_id | 29 | status | 45 | pattern | 61 | cost_budget |
| 14 | sender | 30 | payload | 46 | additionalProperties | 62 | deadline_ms |
| 15 | recipient | 31 | metadata | 47 | content | 63 | retry_count |
| 16 | goal_hash | 32 | headers | 48 | role | 64 | idempotency_key |

in benchmarks/run_benchmarks.py computes a 36-dimensional feature vector $\phi(B_t) = [\phi_0, \phi_1, \ldots, \phi_{35}]^\top \in \mathbb{R}^{36}$ from every observed wire body $B_t \in \{0, \ldots, 255\}^{L_t}$:

$$\begin{aligned}
\phi_0(B_t) &= \frac{L_t \bmod 8193}{8192.0} \quad \text{(normalized frame length)},\\
\phi_1(B_t) &= -\frac{1}{8.0}\sum_{b=0}^{255} \hat{p}(b)\log_2 \hat{p}(b) \quad \text{(normalized entropy)},\\
\phi_2(B_t) &= \sum_{b=32}^{126} \hat{p}(b) \quad \text{(printable ASCII density)},\\
\phi_3(B_t) &= \sum_{c \in \mathcal{D}_{\text{json}}} \hat{p}(\text{ord}(c)) \quad \text{(JSON delimiter density)},\\
\phi_{4+k}(B_t) &= \sum_{j=0}^{7} \hat{p}(8k+j), \quad k \in \{0..31\} \quad \text{(32-bin hist.)},
\end{aligned} \tag{32}$$

where $\mathcal{D}_{\text{json}} = \{\texttt{'\{'}, \texttt{'\}'}, \texttt{':'}, \texttt{'"'}, \texttt{','}\}$. Prior to training, each feature coordinate $d \in \{0, \ldots, 35\}$ is $z$-score standardized using the training-set mean $\mu_d$ and standard deviation $\sigma_d$: $\tilde{\phi}_d = (\phi_d - \mu_d)/(\sigma_d + 10^{-8})$. Across the 800 training traces and 200 stratified test traces (20 classes), the three adversarial classifiers are parameterized as follows:

- **Random-Subspace Nearest-Centroid Ensemble (`Subspace RF`, 50 Estimators):** Draws $B_{\text{tree}} = 50$ bootstrap resamples of size 800 paired with random 12-dimensional feature subspaces $S_b \subset \{0, \ldots, 35\}$ ($|S_b| = 12$), computes class-conditional subspace centroids $\mathbf{c}_{b,y}$ for $y \in \{0, \ldots, 19\}$, and aggregates majority votes across all 50 estimators.
- **2-Layer Multi-Layer Perceptron (MLP):** Parameterizes a feed-forward network $\hat{\mathbf{y}} = \text{Softmax}(\text{ReLU}(\tilde{\boldsymbol{\phi}}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2)$ with input dimension 36, hidden layer width 48, and output dimension 20, trained for 120 full-batch epochs using cross-entropy loss, learning rate $\eta = 0.08$, and $L_2$ weight regularization $\lambda = 10^{-4}$.
- **$k$-Nearest Neighbors ($k$-NN, $k = 5$):** Computes exact Euclidean distances $\|\tilde{\boldsymbol{\phi}}_{\text{test}} - \tilde{\boldsymbol{\phi}}_{\text{train},i}\|_2$ in $\mathbb{R}^{36}$ and predicts the plurality class label among the $k = 5$ nearest training instances.

## APPENDIX G
## PRODUCTION SERVICE-MESH SIDECAR, FASTMCP GATEWAY INTEGRATION, AND OPERATIONAL RESILIENCE

In large-scale enterprise deployments on Google Cloud (Vertex AI Agent Builder, Cloud Run FastMCP servers, and GKE clusters running Anthos Service Mesh), Agent-Chameleon can be deployed in two complementary topologies:

1) **In-Process ADK / FastMCP SDK Middleware:** The `ChameleonMorphEngine` hooks directly into the client's outbound HTTP/2 or gRPC transport interceptor and the server's inbound ASGI/WSGI/gRPC request handler. This topology protects payloads end-to-end across all intermediate L7 proxies, Envoy sidecars, and API gateways, ensuring that even if a co-resident sidecar or service-mesh telemetry collector is compromised, it observes only constant-bucket high-entropy morphed frames ($H(B_t) = 7.992$ bits/byte).
2) **Out-of-Process Confidential VM Sidecar Proxy:** For legacy agent runtimes that cannot modify their SDK dependencies, Agent-Chameleon executes as a co-resident local Unix-domain socket daemon inside the same hardware-attested Confidential VM enclave boundary, morphing outbound JSON-RPC traffic before it exits the enclave virtual NIC.

### *A. Session Bootstrap, Epoch Resynchronization, and Network Partition Recovery*

Table IX summarizes the finite-state machine governing session lifecycle, periodic Tier-2 Ed25519 epoch anchoring, and crash recovery. During initial session establishment ($t = 0$), the initiator transmits its ephemeral X25519 public share $\text{pk}_i$

authenticated under its workload identity certificate or Tier-2 Ed25519 identity key. Once $Z_{\text{shared}} = \text{X25519}(\text{sk}_i, \text{pk}_j)$ is established, all subsequent turns $t \geq 0$ execute with zero asymmetric PKI overhead on the Tier-1 hot path ($28.86\,\mu$s mean ratchet plus Schnorr proof generation). Every $\Delta_{\text{anchor}} = 100$ turns, the sender optionally attaches an out-of-band Tier-2 RFC 8032 Ed25519 signature (`sign_epoch_anchor`) over $(t, R_t)$ in `X-Chameleon-Epoch-Anchor` to provide non-repudiable cross-domain audit checkpoints.

If an underlying TCP/HTTP2 connection drops mid-flight after $\mathcal{A}_i$ seals packet $W_t$ but before $\mathcal{A}_j$ receives it, $\mathcal{A}_i$ simply re-transmits the exact cached wire packet $W_t$ upon transport reconnection. Because $\mathcal{A}_j$ never received $W_t$, its expected epoch remains $t$ and $(t, \tau_{\text{ratchet}}) \notin \mathcal{W}_{\text{seen}}$; thus the idempotent wire retry succeeds on the first attempt. Conversely, if $W_t$ was processed by $\mathcal{A}_j$ and only the HTTP/2 response was lost, $\mathcal{A}_j$ detects the duplicate $(t, \tau_{\text{ratchet}}) \in \mathcal{W}_{\text{seen}}$ and returns a signed epoch synchronization challenge instructing $\mathcal{A}_i$ whether turn $t$ was already committed.

## Appendix H
## Formal Security Reduction of Multi-Epoch Ratchet Indistinguishability and Water-Filling Optimality

**Proposition 3** (Strict Concavity of Shannon Entropy and Optimality of Set-Complement Water-Filling)**.** *Let* $\mathbf{c}^{(0)} = (c_0^{(0)}, \ldots, c_{255}^{(0)}) \in \mathbb{Z}_{\geq 0}^{256}$ *be the byte-frequency histogram of the whitened prefix* $W_p$ *of length* $L_p = \sum_{b=0}^{255} c_b^{(0)}$*, and let* $L_c = B - L_p$ *be the available chaff byte budget. Among all non-negative integer chaff allocations* $\mathbf{\Delta} = (\Delta_0, \ldots, \Delta_{255}) \in \mathbb{Z}_{\geq 0}^{256}$ *satisfying* $\sum_{b=0}^{255} \Delta_b = L_c$*, the plug-in Shannon byte entropy*

$$H(\mathbf{c}^{(0)} + \mathbf{\Delta}) = -\sum_{b=0}^{255} \frac{c_b^{(0)} + \Delta_b}{B} \log_2 \left( \frac{c_b^{(0)} + \Delta_b}{B} \right) \quad (33)$$

*is Schur-concave in the total count vector* $\mathbf{c} = \mathbf{c}^{(0)} + \mathbf{\Delta}$ *and is uniquely maximized (up to permutation of equal-count bins) when* $\mathbf{c}$ *is majorized by every other feasible count vector, i.e.,* $\max_b c_b - \min_b c_b \leq 1$*. Furthermore, the Stage 2B slice-wise set-complement plus* $\mathbb{Z}_{256}$ *affine-bijection construction approximates this Schur-concave optimum in* $\mathcal{O}(B)$ *C-level operations without transmitting an auxiliary histogram side channel.*

*Proof.* The function $\varphi(u) = -(u/B)\log_2(u/B)$ is strictly concave on $u > 0$ since $\varphi''(u) = -\frac{1}{Bu \ln 2} < 0$. By Karamata's inequality (and the theory of Schur-concave functions), a symmetric sum $\sum_{b=0}^{255} \varphi(c_b)$ of strictly concave functions subject to $\sum_{b=0}^{255} c_b = B$ is maximized when the components $c_b$ are as nearly equal as possible. If an algorithm increments the minimum-count bins first ($c_b = 0 \to 1$, then $1 \to 2$, etc.), each discrete transfer of 1 unit from a bin with count $c_{\text{high}} \geq c_{\text{low}} + 2$ to a bin with count $c_{\text{low}}$ strictly increases $H$ by:

$$\Delta H = [\varphi(c_{\text{low}} + 1) - \varphi(c_{\text{low}})] - [\varphi(c_{\text{high}}) - \varphi(c_{\text{high}} - 1)] > 0. \quad (34)$$

In Stage 2B, taking the exact set complement $M^{(s)} = \Sigma_{256} \setminus \text{set}(W_p^{(s)})$ on each 128-byte slice $W_p^{(s)}$ increments every zero-count byte in that slice to 1, eliminating all zero-frequency bins in the first pass, while each subsequent 256-byte affine bijection block $\Pi_{a,r}$ adds the constant vector $\mathbf{1}_{256} = (1, 1, \ldots, 1)$ to $\mathbf{c}$, preserving the bounded variance $\mathbb{E}[\sum_b (c_b - B/256)^2] \leq L_p + 64$ while scaling $B = 256M$ and driving $H(B_t) \to 8.000$ bits/byte at rate $8 - \mathcal{O}(1/M)$. □

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TABLE IX
OPERATIONAL STATE MACHINE, FAILURE MODES, AND CRYPTOGRAPHIC RECOVERY GUARANTEES OF AGENT-CHAMELEON

| **Lifecycle Phase / Event** | **Trigger Condition** | **Cryptographic Action Executed** | **Latency / Security Invariant** |
|---|---|---|---|
| 1. Session Bootstrap | New A2A / MCP channel open | Ephemeral X25519 ECDH + HKDF root $R_{-1}$ init | 1-RTT handshake; forward-secure root |
| 2. Hot-Path Turn ($t \geq 0$) | Outbound tool call $P_t$ under $\mathcal{G}_t$ | Advance $R_t$, Feistel $F_k$, Codec $m_t$, Chaff, Schnorr $\pi_t$ | 28.86 $\mu$s mean (Ratchet + ZK Prove) |
| 3. Periodic Anchor ($t \equiv 0 \bmod 100$) | Epoch checkpoint boundary | Sign $\text{Ed25519}_{\text{sk}}(\texttt{"anchor"} \parallel t \parallel R_t)$ | 179.60 $\mu$s amortized ($+1.80$ $\mu$s/turn) |
| 4. Forged / Bit-Flipped Packet | $\tau_{\text{ratchet}}$ or $g^{s_t} \neq R_{\text{com}} Y_t^{e_t}$ | Raise `PermissionError`; restore $(R_{t-1}, t, D_{t-1})$ | 100% reject; 0 state drift on receiver |
| 5. Replayed Packet ($t' \leq t$) | $(t', \tau_{\text{ratchet}}) \in \mathcal{W}_{\text{seen}}$ or $t' \neq t$ | Reject in $\mathcal{O}(1)$ hash lookup prior to codec parse | 100% replay immunity; DoS-resistant |
| 6. Goal Context Mismatch | Caller $\mathcal{G}_t \neq$ Receiver $\mathcal{G}'_t$ | Divergent $H(\mathcal{G}_t) \implies R_t \neq R'_t \implies Y_t \neq \hat{Y}_t$ | 100% cross-goal tool hijacking block |


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